

Supplementary Material

Forward KL Can Be Better: Normalizing Flow Boltzmann Sampling with X-Regularization

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This supplement collects the full derivations and experiments abridged in the main text: the Fisher–Rao gradient-flow analysis (Section [S1](#)), the model-problem study (Section [S2](#)), and the molecular potential-energy experiments (Section [S3](#)). Cross-references of the form (1), Figure 1, Theorem 1, Section 3 point to the main text; those of the form (S1), Figure S1 are internal to this supplement.

S1 Fisher–Rao gradient flow with X functionals

Before turning to the training algorithm, we study the gradient-flow properties of the loss on the space of densities; the convergence rate of this continuous-time idealization is a proxy for how trainable the corresponding discrete loss is in practice. We view ν as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities ([Ambrosio et al., 2008](#); [Villani, 2009](#)) and ask how fast ν_t converges to the target μ in $\text{KL}(\mu\|\nu_t)$.

In this section we restrict to the loss with only the forward KL and the shape regularizer X_μ ,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (\text{S1})$$

where $\lambda > 0$ is a fixed constant. Throughout this section we abbreviate the importance weight by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is $z(x) = \log w(x) = \log \mu(x) - \log \nu(x)$, the field z of [\(5\)](#). The importance weight w is distinct from the localization measure ω in X_ω .

The choice of Riemannian metric matters. In the Wasserstein-2 geometry, induced by optimal transport ([Jordan et al., 1998](#); [Otto, 2001](#)), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani / Bakry–Émery machinery ([Otto and Villani, 2000](#); [Bakry et al., 2014](#)) that yields exponential convergence for the reverse KL does not apply. We therefore work in the Fisher–Rao (FR) geometry ([Amari, 2016](#); [Bauer et al., 2016](#)), in which the forward KL admits an unconditional exponential decay rate and the regularizer X_μ contributes a qualitatively new, nonlocal birth–death modulation ([Lu et al., 2019](#)): the local update at x is driven by how the importance weight at x ranks against the population of weights under μ .

Denoting the ν -mean by $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$, the Fisher–Rao gradient of a functional $\mathcal{F}[\nu]$ and the associated dissipation identity along the FR gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_{\nu_t} \right)^2 \nu_t dx. \quad (\text{S2})$$

The mean-subtraction enforces $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the FR flow conserves probability mass.

S1.1 Derivation of X_μ -regularized Fisher–Rao gradient flow

The first variation of the forward KL with respect to ν admits the form

$$\frac{\delta \text{KL}(\mu \parallel \nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (\text{S3})$$

so by (S2), using $\langle w \rangle_\nu = \int_{\mathbb{R}^d} \mu \, dx = 1$, the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}}[\text{KL}(\mu \parallel \nu)](x) = \nu(x) - \mu(x). \quad (\text{S4})$$

For the X_μ functional, the inner log-ratio difference has first variation

$$\frac{\delta(\log w(x) - \log w(y))}{\delta \nu(x)} = \frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)}.$$

By the symmetry of X_μ in x and y ,

$$\begin{aligned} \frac{\delta X_\mu(\mu \parallel \nu)}{\delta \nu}(x) &= 2 \int_{\mathbb{R}^d} \mu(x) \mu(y) \text{sgn}(w(x) - w(y)) \frac{\delta(\log w(x) - \log w(y))}{\delta \nu(x)} dy \\ &= 2 \int_{\mathbb{R}^d} \mu(x) \mu(y) \text{sgn}(w(x) - w(y)) \left(\frac{1}{\nu(y)} \delta(x - y) - \frac{1}{\nu(x)} \right) dy \\ &= -2w(x) \int_{\mathbb{R}^d} \mu(y) \text{sgn}(w(x) - w(y)) dy, \end{aligned}$$

where the contribution of $\delta(x - y)$ vanishes because $\text{sgn}(w(x) - w(y)) = 0$ at $y = x$. Hence

$$\frac{\delta X_\mu(\mu \parallel \nu)}{\delta \nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y \sim \mu}[\text{sgn}(w(x) - w(y))]. \quad (\text{S5})$$

We call S the *rank field* of the importance weight under μ : $S(x) > 0$ where $w(x)$ exceeds the μ -median of w , i.e. where ν undercovers a mode of μ ; $S(x) < 0$ where ν overcovers; and $S \equiv 0$ at $\nu = \mu$. By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x, y \sim \mu}[\text{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of X_μ with respect to ν is

$$\text{grad}^{\text{FR}}[X_\mu(\mu \parallel \nu)](x) = -2\mu(x)S(x). \quad (\text{S6})$$

Since the FR gradient is linear in the functional, the FR gradient of $\mathcal{L}[\nu] = \text{KL}(\mu \parallel \nu) + \lambda X_\mu(\mu \parallel \nu)$ is the direct sum of (S4) and (S6):

$$\text{grad}^{\text{FR}}[\mathcal{L}](x) = \nu(x) - \mu(x) - 2\lambda\mu(x)S(x).$$

The regularized Fisher–Rao gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}}[\mathcal{L}]$ therefore reads

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda\mu(x)S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[\text{sgn} \left(\frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (\text{S7})$$

Equation (S7) is a *nonlocal* PDE: its right-hand side at x depends on $S_t(x)$, which in turn requires the entire profile of the importance weight μ/ν_t over the support of μ . Mass is conserved along the flow because

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 2\lambda \int_{\mathbb{R}^d} \mu(x) S_t(x) dx = 0,$$

and the flow is stationary at $\nu_t = \mu$, where $\mu(x)/\nu_t(x) \equiv 1$ and hence $S_t \equiv 0$. When $\lambda = 0$, (S7) reduces to the pure Fisher–Rao gradient flow of the forward KL.

S1.2 Convergence analysis of the X_μ -regularized flow

To analyze the convergence rate of the regularized flow (S7), we take the forward KL $\text{KL}(\mu\|\nu_t)$ as the error metric and differentiate it along the flow:

$$\frac{d}{dt}\text{KL}(\mu\|\nu_t) = -\int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx,$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral is settled by the following rank-correlation identity.

Lemma S1 *For any probability density ν on $\mathbb{R}^d$,*

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|] \geq 0,$$

with equality if and only if w is μ -almost-everywhere constant.

Proof. Since $\mu^2/\nu = \mu w$, the definition of S gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_\mu [w(x) S(x)] = \mathbb{E}_{x,y \sim \mu} [w(x) \text{sgn}(w(x) - w(y))].$$

Symmetrizing in $x \leftrightarrow y$ gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [(w(x) - w(y)) \text{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y \sim \mu} [|w(x) - w(y)|].$$

□

Applying Lemma S1 with $\nu = \nu_t$ and $S = S_t$, the dissipation rewrites as

$$\frac{d}{dt}\text{KL}(\mu\|\nu_t) = -\chi^2(\mu\|\nu_t) - \lambda \mathbb{E}_{x,y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (\text{S8})$$

The elementary inequality $w \log w \leq w^2 - w$ applied to $w = \mu/\nu_t$ gives $\chi^2(\mu\|\nu_t) \geq \text{KL}(\mu\|\nu_t)$, hence $\frac{d}{dt}\text{KL}(\mu\|\nu_t) \leq -\text{KL}(\mu\|\nu_t)$, and Grönwall's lemma yields the exponential decay

$$\text{KL}(\mu\|\nu_t) \leq e^{-t} \text{KL}(\mu\|\nu_0), \quad t \geq 0, \quad (\text{S9})$$

with no structural assumption on μ : neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

The uniform convergence rate (S9) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR flow is a continuous-time idealization on the space of densities and is agnostic to the sampling cost. In practice, the difficulty of training has been partially transported to the AIS step: evaluating the forward KL and its gradient requires accurate samples of μ , and producing such samples in high dimensions is itself a hard problem. This is the reason that AIS is needed, and that the $X_{\hat{\mu}}$ regularizer must compensate for modes that AIS alone fails to reach. The exponential rate (S9) quantifies how fast the flow *can* converge given accurate target samples; it does not eliminate the burden of producing them.

A naive comparison of dissipation rates between KL-only and $\text{KL} + \lambda X_\mu$ would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of X_μ is structural rather than rate-based: while the pure KL FR gradient is local in x , the rank field S_t is global, so the regularized flow (S7) couples the update at x to the entire distribution of

importance weights under μ . This nonlocal coupling amplifies the birth rate on undercovered regions of μ and suppresses it on overcovered ones, a global comparison that pure forward KL, which knows only whether $\mu(x)/\nu_t(x)$ exceeds or falls below 1 pointwise, cannot produce.

Two further cautions concern the discrete realization of the flow. First, the time-reparametrization argument cuts in both directions once the optimizer enters. Training follows Adam steps, not the FR flow: the per-coordinate second-moment normalization absorbs a global rescaling of the loss, so a larger gradient buys no speedup by itself, while a regularizer that reshapes the relative magnitudes of the gradient components does change the optimizer’s effective time scale. Part of any observed difference in training speed between KL and $\text{KL} + \lambda X_\mu$ may therefore be optimizer scaling rather than information geometry, and we claim no faster transient on this basis; the rejected-rung values of Section S2.5 confirm that none materializes. Second, the dissipation (S8) is a population statement, while each gradient step sees an estimate from a batch of B samples. At large batch the estimated forward KL gradient already approximates its population direction well, the idealized rate (S9) is nearly realized by the bare loss, and the headroom left for the X terms narrows; the regularizer earns its keep when the training signal is noisy or biased: small batches, early bridges, imperfect AIS samples. The clock ladder of Table S6 shows this contraction directly: the early-rung validation ESS gap shrinks from +0.21 at $B = 10^3$ to +0.13 at $B = 10^5$.

Remark (Monotone decay of X_μ) *The regularizer X_μ also decreases monotonically along the regularized flow: pairing the variation $\delta X_\mu/\delta\nu = -2wS$ (S5) against the flow (S7) gives*

$$\frac{d}{dt} X_\mu(\mu\|\nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0,$$

with strict inequality unless μ/ν_t is μ -almost-everywhere constant, the first term being non-positive by Lemma S1 and vanishing only in that case. Thus X_μ is itself driven to zero in the long-time limit.

S1.3 Asymptotic accuracy under biased AIS samples

The convergence analysis of Section S1.2 assumes that the outer expectation in $\text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is taken under the exact target μ . In practice, the algorithm only has access to an approximation $\tilde{\mu}$ of μ supplied by AIS, and the realized loss reads

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (\text{S10})$$

where the pointwise ratio $w(x) = \mu(x)/\nu(x)$ is still evaluated against the exact target, since it is computed per sample from the target potential U and the flow’s own density. Repeating the derivation of Section S1.1 with μ replaced by $\tilde{\mu}$ in the outer integrands gives the Fisher–Rao gradient

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda \tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}} [\text{sgn}(w(x) - w(y))].$$

A stationary point ν^* of (S10) therefore satisfies the self-consistent equation

$$\nu^*(x) = \tilde{\mu}(x) \left[1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) dy \right]. \quad (\text{S11})$$

At $\lambda = 0$ the equation collapses to $\nu^* = \tilde{\mu}$: pure forward KL training inherits the bias of the AIS surrogate $\tilde{\mu}$ verbatim, and the residual error is exactly $\text{KL}(\mu\|\tilde{\mu})$. For $\lambda > 0$, however, the rank-based correction $1 + 2\lambda \tilde{S}^*$, where $\tilde{S}^*$ denotes $\tilde{S}$ evaluated at $\nu = \nu^*$, inflates ν^* above $\tilde{\mu}$ where the importance weight μ/ν^* is high, that is, where ν^* still undercovers μ , and deflates it where the weight is low, providing a global feedback that pulls ν^* from $\tilde{\mu}$ back toward μ . Two facts prepare the statement. For $\lambda \leq \frac{1}{2}$ the bracket in (S11) is non-negative, because $\tilde{S}^* \in [-1, 1]$. Moreover, the antisymmetry of the sign in x, y implies

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \tilde{S}^*(x) dx = \mathbb{E}_{x, y \sim \tilde{\mu}} \left[\text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) \right] = 0,$$

hence every stationary point $\nu^* = \tilde{\mu}(1 + 2\lambda \tilde{S}^*)$ is a probability density.

Theorem S1 Assume that μ is absolutely continuous with respect to $\tilde{\mu}$ and that $\lambda \in [0, \frac{1}{2}]$. Then every stationary density ν^* of the X -regularized loss $\tilde{\mathcal{L}}$ (S10) satisfies

$$\text{KL}(\mu\|\nu^*) \leq \text{KL}(\mu\|\tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x)\tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (\text{S12})$$

Proof. Substituting the stationary identity $\nu^* = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)$ into the difference of forward KLs and using $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)(\mu/\nu^*)$,

$$\text{KL}(\mu\|\tilde{\mu}) - \text{KL}(\mu\|\nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda\tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} u(x) \log u(x) dx,$$

where $u(x) := 1 + 2\lambda\tilde{S}^*(x) \geq 1 - 2\lambda \geq 0$, since $\tilde{S}^* \in [-1, 1]$ (a $\tilde{\mu}$ -average of signs) and $\lambda \leq \frac{1}{2}$. The elementary inequality $u \log u \geq u - 1$ for $u \geq 0$ then yields

$$\text{KL}(\mu\|\tilde{\mu}) - \text{KL}(\mu\|\nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

Applying the rank-correlation argument of Lemma S1 verbatim with μ replaced by $\tilde{\mu}$ in the outer measure,

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x)\tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (S12). $\square$

The right-hand side of (S12) (Theorem S1) contrasts the two roles of X_μ training. At $\lambda = 0$ the bound is tight at $\nu^* = \tilde{\mu}$ and the residual KL is fixed at $\text{KL}(\mu\|\tilde{\mu})$; the AIS bias is a hard floor. Switching on the regularizer with $\lambda \leq \frac{1}{2}$ lowers this floor (strictly, unless μ/ν^* is constant on the support of $\tilde{\mu}$) by an amount controlled by the discrepancy of the importance weights μ/ν^* across the support of $\tilde{\mu}$: the larger this discrepancy, the more the rank-based feedback pulls ν^* toward μ . The X_μ regularizer is therefore not merely a numerical stabilizer for the forward KL flow: it lowers the asymptotic accuracy floor of the trained density when the target samples are biased, which is the practically relevant regime.

The threshold $\lambda \leq \frac{1}{2}$ is a worst-case condition enforcing positivity of ν^* uniformly in x ; in practice the accuracy improvement persists well beyond it. The practical takeaway is that even without the wide-coverage regularizer $X_{\tilde{\mu}}$, adding X_μ alone to the forward KL loss already reduces the bias floor inherited from inaccurate AIS samples.

S2 Numerical experiments on model problems

We validate the X -regularized losses on a suite of model targets; the application to a real molecular system is taken up in Section S3. The suite spans three families: multimodal probability distributions, the six 2D benchmarks (Section S2.1) and the high-dimensional product multi-well (Section S2.3); lattice field theories, the tilted ϕ^4 theory (Section S2.4) and the p -state clock model (Section S2.5); and Bayesian inverse problems, sensor-array source localization (Section S2.2) and screened Poisson source inversion (Section S2.6). Throughout this section, vectors are set in bold ($\mathbf{x}$, $\mathbf{y}$, $\boldsymbol{\theta}$, $\boldsymbol{\phi}$) and their scalar components and scalar quantities upright.

S2.1 2D benchmark distributions

We test the training scheme on six 2D targets, each chosen to isolate a different failure mode of the forward KL rather than to match the Gaussian mixture and double-well community suite (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024): Threewell, Himmelblau, Annulus, Python, Chessboard, and Sparse. The potentials and mode structures are listed in Table S1; each target’s construction is described alongside its results below.

For all six targets the source μ_0 is an isotropic Gaussian $\mathcal{N}(0, \sigma^2 I)$ with σ between 0.25 and 2.0; on Threewell it is deliberately narrow and centered at the origin, far from all three wells, so the initial push-forward covers none of them. The flow is a neural spline flow (NSF) (Durkan et al., 2019) with 6 coupling

name	potential $U(x_1, x_2)$	minima
Threewell	$6[(x_1^2 - 1)^2 + (x_2^2 - 1)^2 + \sin(x_1 + 2x_2)]$	three asymmetric wells
Himmelblau	$(x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$	four isolated points
Annulus	$20(r^6 - 8r^4 + 16r^2 + 1)^{\frac{1}{3}}, r^2 = x_1^2 + x_2^2$	$r = 0$ and ring at $r = 2$
Python	Python-like Gaussian mixture	1024 stroke centers + 2 eye centers
Chessboard	Chessboard-like Gaussian mixture	26 of 32 black cells, 6 removed
Sparse	Sparse Gaussian mixture	2 dense + 2 isolated centers

Table S1: Six 2D benchmark targets.

transforms, 32 bins, and hidden widths (128, 128), on a square box matched to each target. Training uses Adam at learning rates 10^{-3} to 5×10^{-3} for 1000 steps, 2000 on Python whose narrow tubular support takes longer to saturate, with batch size 500, enlarged to 1000–2000 on the multimodal Threewell, Chessboard, and Sparse so a single batch can cover all modes. At each step the AIS surrogate is one importance-weighted resampling pass against the current pushforward followed by 50 Langevin steps. The loss coefficients are the balanced hyperparameters of Section 3.1, $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$, and the wide-coverage measure $\hat{\mu}$ is built once before training by QT (Algorithm 2) and recycled with one Langevin rejuvenation per step.

For each target we compare four losses obtained by progressively adding one X functional at a time: the bare forward KL (2); KL+ X_μ with $\lambda = 1$; KL+ X_μ + $X_{\hat{\mu}}$ with $\alpha = 1$, $\beta = 0$; and the balanced mixture KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$ with $\alpha = \beta = \frac{1}{2}$.

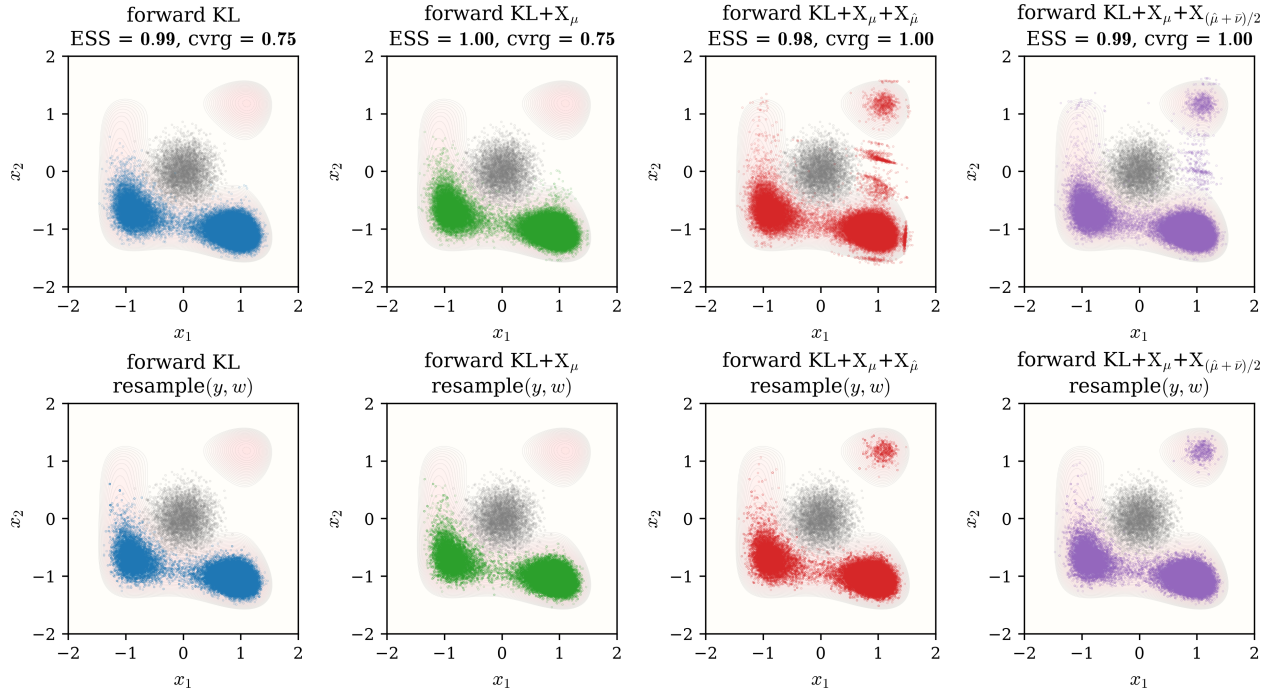
Threewell. The quartic-plus-sinusoid potential has three asymmetric wells. Forward KL and KL+ X_μ lock onto only two of them and never discover the third, yet their ESS stays close to one: a fake ESS instance, the flow omitting a mode while reporting near-uniform weights on the support it reaches. Adding $X_{\hat{\mu}}$ places mass on the QT-discovered third well, and the mixture attains the same full coverage at the highest full-coverage ESS (Figure S1a).

Himmelblau. Forward KL and KL+ X_μ both cover only two of the four modes, yet the ESS gives no warning: KL+ X_μ posts the highest nominal ESS of the four while missing half the modes, a fake ESS instance. Adding $X_{\hat{\mu}}$ or the mixture recovers full coverage and ties at the best full-coverage ESS (Figure S1b).

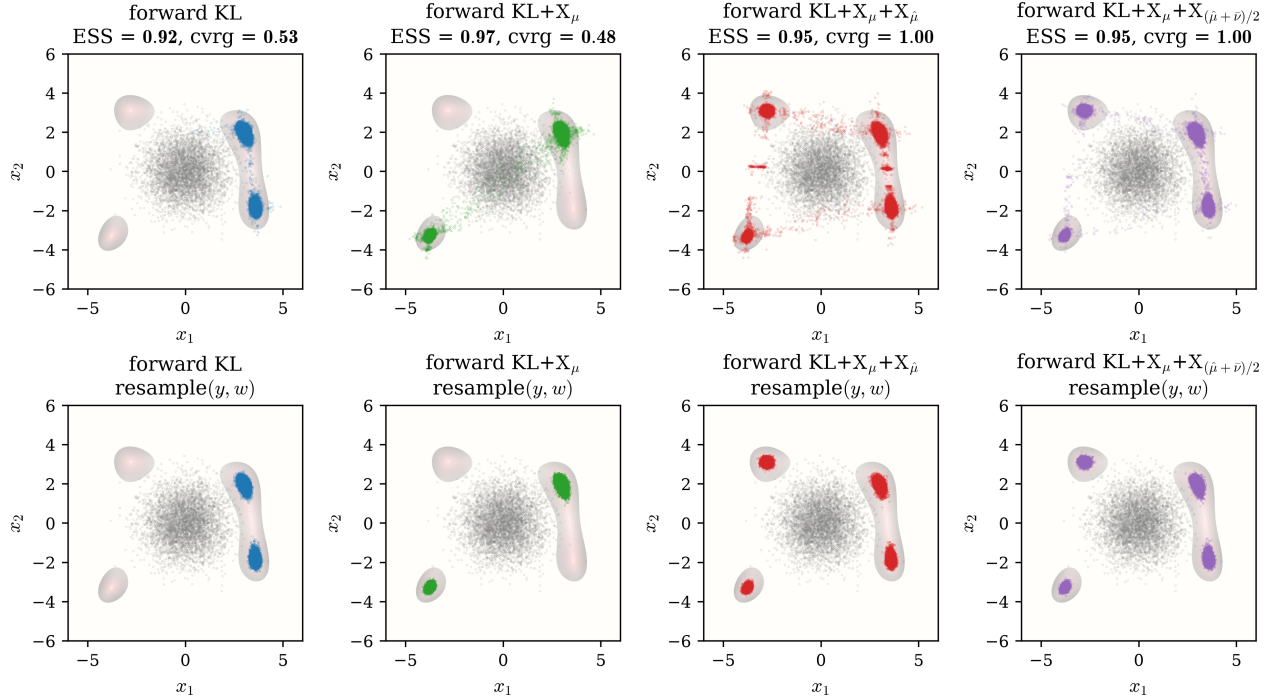
Annulus. All four methods reach full coverage, so the discriminator is the flow shape and ESS. Forward KL alone leaves visible axis-aligned filaments across the central barrier; adding X_μ tightens the pushforward to a clean ring-plus-point geometry and sharply lifts the ESS, and the mixture ties KL+ X_μ at the best ESS of the four (Figure S2a). This isolates the role of X_μ as a shape regularizer, penalizing pointwise log-ratio variation that the global forward KL leaves untouched.

Python. The Python target lives on an essentially one-dimensional manifold, the skeleton of the two interlocking snake bodies of the Python logo. All four methods reach full coverage and the ESS plateaus well below 1. No mode discovery is needed here, yet the X-regularized losses still improve on bare forward KL, and the mixture reaches the best ESS of the four, leading the next-best X variant by ≈ 0.10 (Figure S2b) — consistent with $\bar{\nu}$ anchoring the off-manifold profile of ν .

Chessboard. Coverage is full for all four methods, but mode structure and ESS vary. Forward KL leaks mass into the six removed cells, and neither X_μ nor $X_{\hat{\mu}}$ stops the leakage: $\hat{\mu}$ alone is sample-blind to the removed cells. The mixture resolves it: the $\bar{\nu}$ half sees the holes whenever ν leaks mass there, and the cross contributions $|\log(\mu/\nu)(\mathbf{y}) - \log(\mu/\nu)(\mathbf{y}')|$ with $\mathbf{y} \sim \hat{\mu}$ and $\mathbf{y}' \sim \bar{\nu}$ anchor in-cell and out-of-cell log-ratios to each other, placing mass on the filled cells at the best ESS (Figure S3).

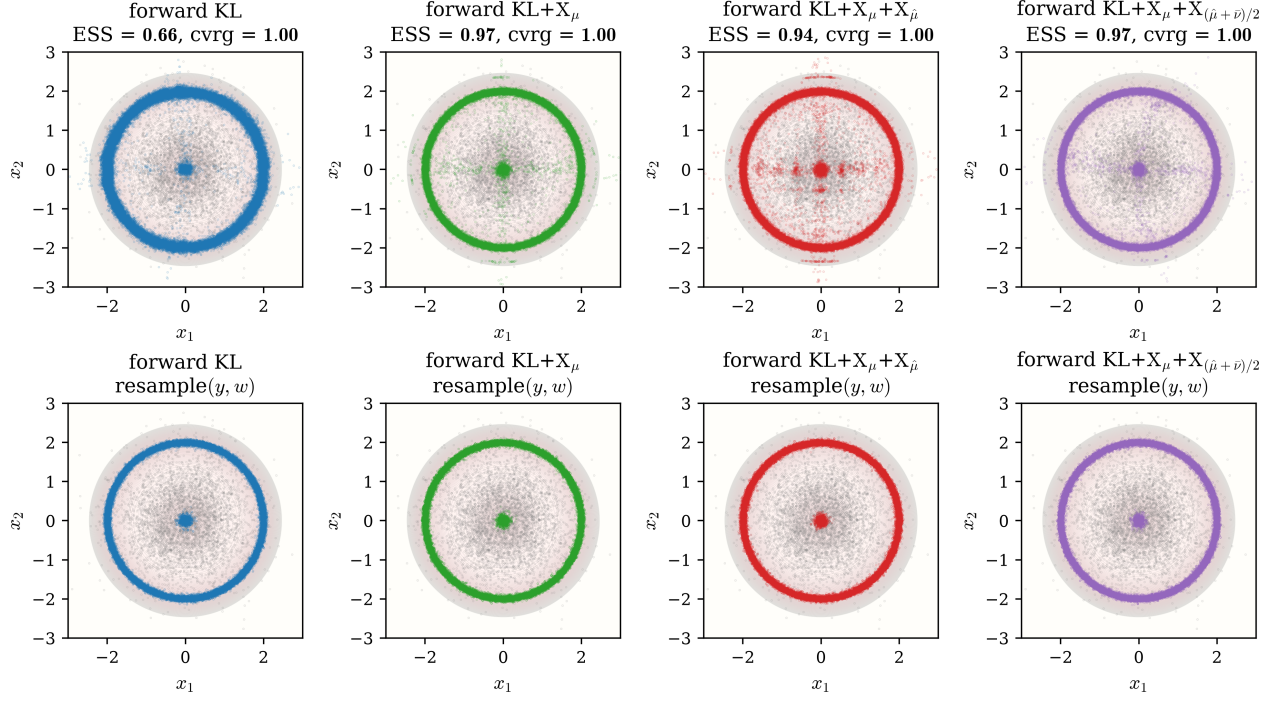


(a) Threewell target.

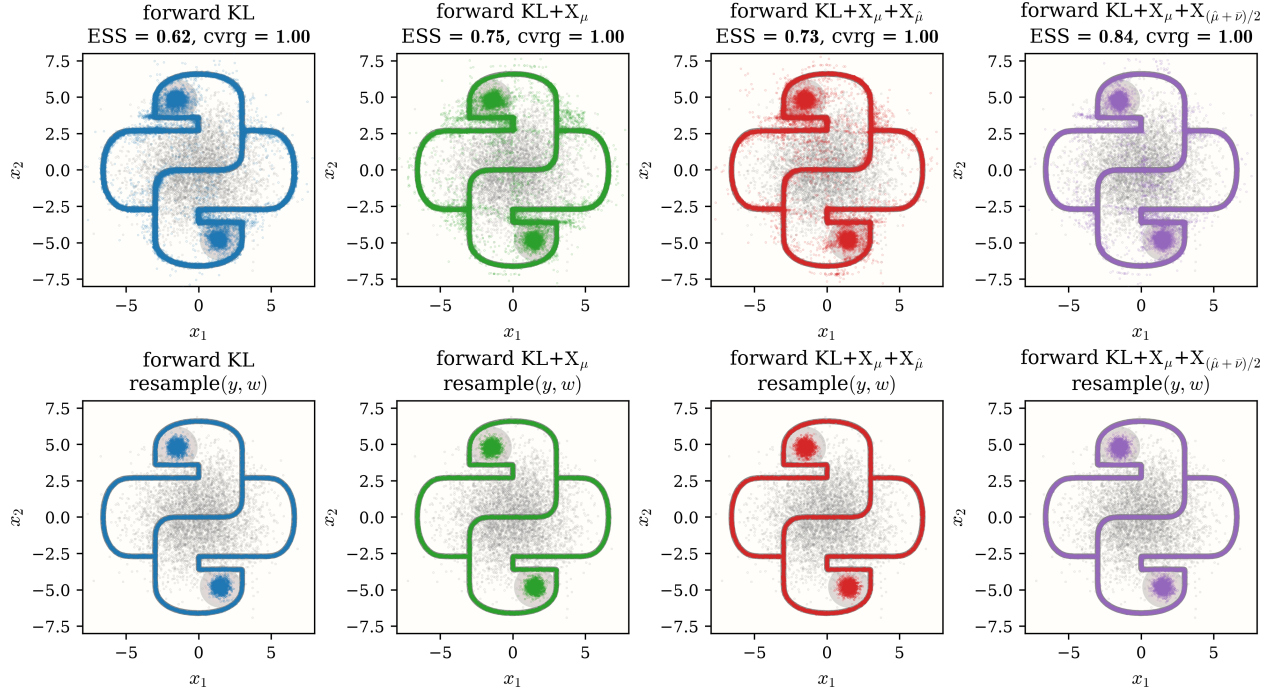


(b) Himmelblau target.

Figure S1: Mode discovery targets. In each panel, Row 1 is the pushforward $\mathbf{y} = G^{-1}(\mathbf{x})$ and Row 2 is $\text{resample}(\mathbf{y}, w)$ with Langevin rejuvenation. Columns left to right: forward KL, KL+ X_{μ} , KL+ $X_{\mu}+X_{\hat{\mu}}$, KL+ $X_{\mu}+X_{(\hat{\mu}+\hat{\nu})/2}$. Each pushforward title reports the final ESS and the $\kappa = 5$ coverage against $\hat{\mu}$.



(a) Annulus target.



(b) Python target.

Figure S2: Shape and manifold targets. Rows and columns as in Figure S1.

Sparse. A four-mode Gaussian mixture with a dense central pair and two isolated corner modes the narrow source cannot reach. Forward KL and $\text{KL} + \text{X}_\mu$ lock onto the central pair and miss the corners at high, stable ESS, another fake ESS instance. Adding $\text{X}_{\hat{\mu}}$ restores full coverage but, lacking the $\bar{\nu}$ half, overfits $\hat{\mu}$ and leaves artificial bridges of mass between the modes; the mixture also reaches full coverage while removing these bridges, at the best full-coverage ESS (Figure S4).

target	KL	KL+ X_μ	KL+ $\text{X}_\mu + \text{X}_{\hat{\mu}}$	KL+ $\text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$
Threewell	0.99 (0.75)	1.00 (0.75)	0.98 (1.00)	0.99 (1.00)
Himmelblau	0.92 (0.53)	0.97 (0.48)	0.95 (1.00)	0.95 (1.00)
Annulus	0.66 (1.00)	0.97 (1.00)	0.94 (1.00)	0.97 (1.00)
Python	0.62 (1.00)	0.75 (1.00)	0.73 (1.00)	0.84 (1.00)
Chessboard	0.77 (1.00)	0.76 (1.00)	0.78 (1.00)	0.84 (1.00)
Sparse	0.88 (0.50)	0.94 (0.50)	0.87 (1.00)	0.93 (1.00)

Table S2: Final ESS, with $\kappa = 5$ coverage against $\hat{\mu}$ in parentheses, for each loss on each 2D target; the best full-coverage ESS per row is in bold.

Across the six targets two regimes emerge (Table S2). On the mode discovery targets Threewell, Himmelblau, and Sparse, the ESS stays high for every loss and is therefore useless on its own: forward KL and $\text{KL} + \text{X}_\mu$ post ESS up to 1.00 while silently missing modes, and only the $\text{X}_{\hat{\mu}}$ and mixture variants restore full coverage. On the shape, manifold, and leakage targets Annulus, Python, and Chessboard, coverage is already complete and ESS becomes the discriminator: X_μ helps on Annulus and Python but is flat on Chessboard, whereas the mixture improves on bare forward KL by +0.31, +0.22, and +0.07 and attains the best ESS on all three. Overall the mixture is the only loss of the four that reaches full mode coverage and the best or tied-best full-coverage ESS on every target, and its ESS curves are smoother than bare forward KL.

The mixture loss $\mathcal{L} = \text{KL}(\mu \parallel \nu) + \text{X}_\mu(\mu \parallel \nu) + \text{X}_{(\hat{\mu} + \bar{\nu})/2}(\mu \parallel \nu)$ with $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$ is the recommended default. It absorbs the mode discovery role of $\text{X}_{\hat{\mu}}$ while suppressing the leakage that $\text{X}_{\hat{\mu}}$ alone can introduce when the target has holes interleaved with its modes.

S2.2 Bayesian inverse problem: sensor-array source localization

Our next benchmark is a physical Bayesian inverse problem where the multimodality comes from a symmetry of the measurement, not from a hand-built mixture. The setting is source localization, a standard problem in acoustics and signal processing: a line of fixed sensors records the combined signal of several identical point sources, and we infer the source positions from the noisy readings. The unknown is the vector of source positions $\boldsymbol{\theta} = (\theta_1, \dots, \theta_n) \in \mathbb{R}^n$ for $n = 3$ sources. A source at θ_i leaves a Gaussian bump of width $\ell = 0.8$ at each sensor and the sensors read the sum, so with $J = 25$ sensors at fixed positions $s_1, \dots, s_J$ equispaced on $[-6, 6]$ the reading at sensor j is

$$g_j(\boldsymbol{\theta}) = \sum_{i=1}^n \exp\left(-\frac{(s_j - \theta_i)^2}{2\ell^2}\right).$$

The data are these readings at the true positions $\boldsymbol{\theta}^* = (5, 0, -5)$ with i.i.d. Gaussian noise,

$$y_j = g_j(\boldsymbol{\theta}^*) + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, \sigma_{\text{obs}}^2), \quad \sigma_{\text{obs}} = 0.3,$$

fixed across all losses. The prior on the positions is a centered isotropic Gaussian $\mu_0 = \mathcal{N}(0, \sigma_{\text{prior}}^2 I_n)$ with $\sigma_{\text{prior}} = 1.6$, which is also the flow’s source, and the negative log-posterior is the prior plus the data fit,

$$U(\boldsymbol{\theta}) = \frac{1}{2\sigma_{\text{obs}}^2} \sum_{j=1}^J (g_j(\boldsymbol{\theta}) - y_j)^2 + \frac{1}{2\sigma_{\text{prior}}^2} \|\boldsymbol{\theta}\|^2. \quad (\text{S13})$$

Because the sources are identical, g_j is a sum over them and is unchanged by any permutation of the coordinates, $g_j(\theta_{\pi(1)}, \dots, \theta_{\pi(n)}) = g_j(\boldsymbol{\theta})$ for $\pi \in S_3$; the prior is permutation-invariant too, so $U(\pi \cdot \boldsymbol{\theta}) = U(\boldsymbol{\theta})$.

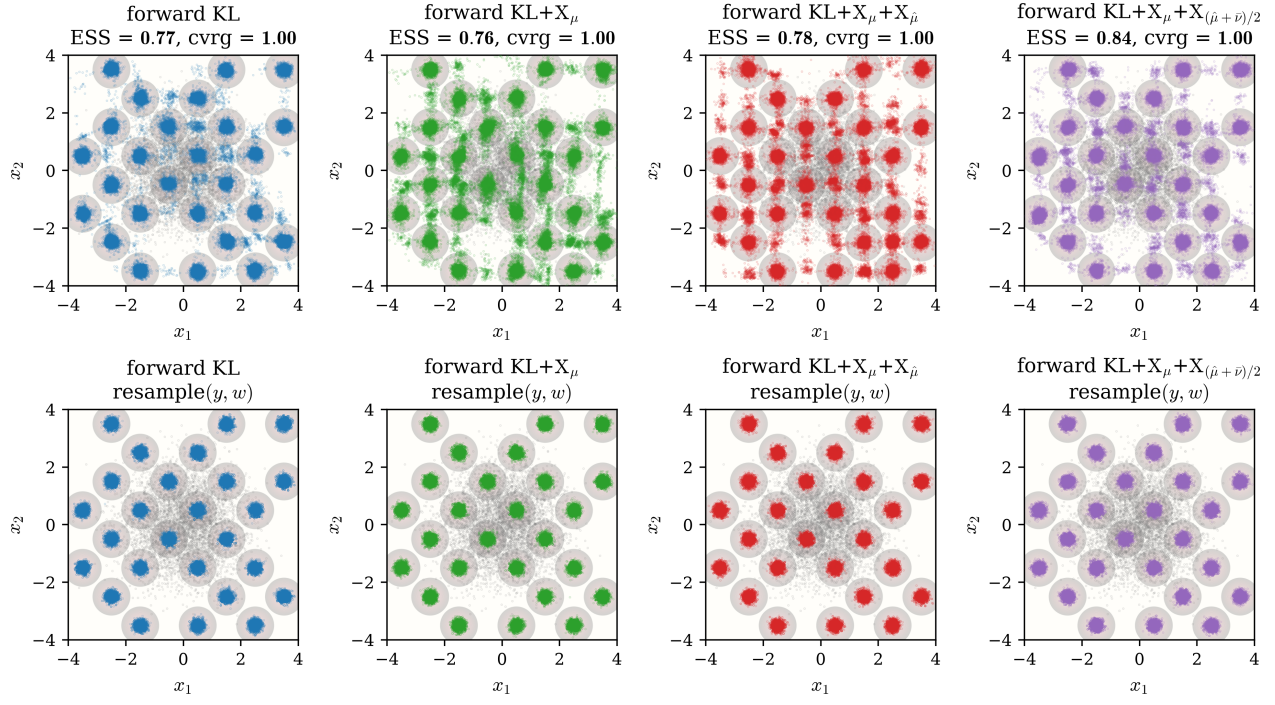


Figure S3: Chessboard target. Row 1: pushforward $\mathbf{y} = G^{-1}(\mathbf{x})$. Row 2: $\text{resample}(\mathbf{y}, w)$ with Langevin rejuvenation. Columns as in Figure S1.

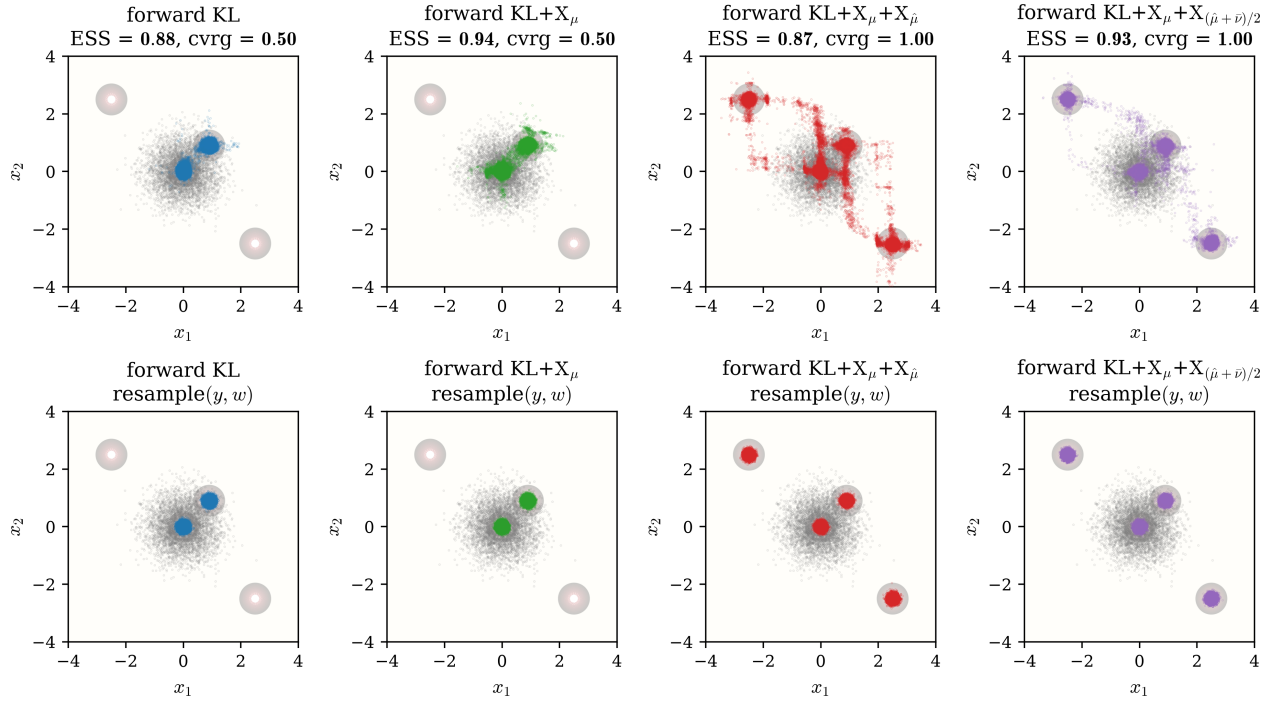


Figure S4: Sparse target. Row 1: pushforward $\mathbf{y} = G^{-1}(\mathbf{x})$. Row 2: $\text{resample}(\mathbf{y}, w)$ with Langevin rejuvenation. Columns as in Figure S1.

The posterior therefore has $3! = 6$ equal, well-separated modes, the permutations of θ^* , and the centered prior sits on their common symmetry point at the origin, about 4.4 prior standard deviations from every mode, $\|\theta^*\| = 7.07$. The prior tail reaches none of them, so a flow trained by bare forward KL only sees the one or two modes its pushforward first lands on and collapses there. We use the same flow, training, and diagnostics as Section S2.1.

Table S3 and Figure S5 report the comparison. Bare forward KL covers only 2 of the 6 modes at ESS 0.90, and adding X_μ posts the highest ESS of the four, 0.98, while collapsing onto a single mode — the ESS improves as coverage gets worse. The QT-driven $X_{\hat{\mu}}$ and the mixture both recover all 6 modes, with mass split evenly, at ESS 0.91 and 0.93.

loss	final ESS	modes found
forward KL	0.904	2
KL+ X_μ	0.979	1
KL+ X_μ + $X_{\hat{\mu}}$	0.913	6
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.934	6

Table S3: Sensor-array source-localization benchmark (S13) with $n = 3$ sources and $3! = 6$ permutation modes.

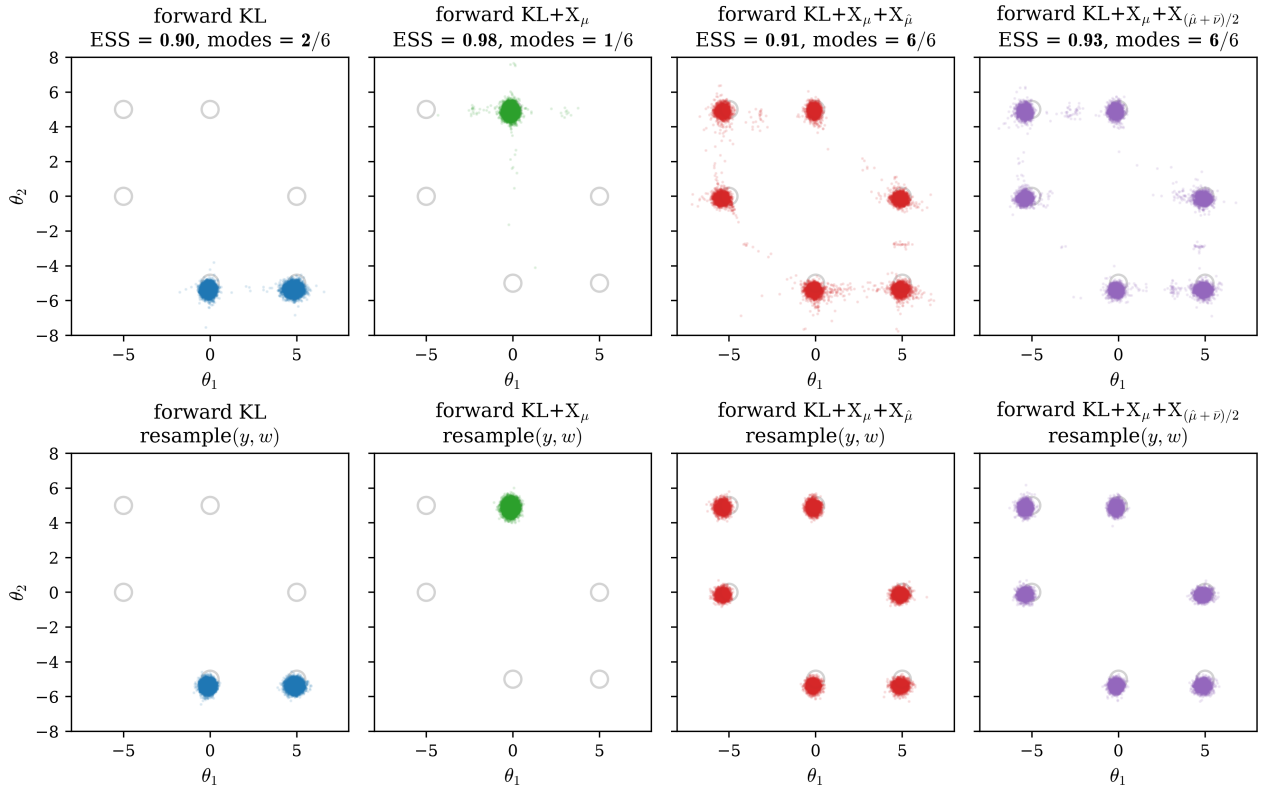


Figure S5: Sensor-array posterior, pushforward positions projected onto (θ_1, θ_2) . Row 1: raw pushforward $\mathbf{y} = G^{-1}(\mathbf{x})$; Row 2: $\text{resample}(\mathbf{y}, w)$ with Langevin rejuvenation. Columns left to right: forward KL, KL+ X_μ , KL+ X_μ + $X_{\hat{\mu}}$, KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$. Light-gray circles mark the six permutation mode centers; each title reports the final ESS and the modes found out of six.

This benchmark is intentionally low-dimensional, and the numbers come from a single training seed; which modes a collapsed method happens to keep is set by the random path, not by the loss.

S2.3 High-dimensional product multi-well

To test the framework beyond two dimensions we construct a family of product multi-well targets in dimension $d = 2^k$, playing here the role that double-well systems play in the diffusion-sampler literature (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024), but with an analytically known mode structure and a diagonal optimal transport map so that training failures are unambiguously training failures and not capacity failures of the flow,

$$U(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^{2^k} x_i^2 + 12 \sum_{i=1}^k \exp(-x_i^2), \quad (\text{S14})$$

a double well in each of the first k coordinates, with minima at $\pm\sqrt{\ln 24} \approx \pm 1.78$ separated by a barrier of height ≈ 9.9 , and a standard Gaussian in the remaining $2^k - k$. The source is the standard Gaussian $\mu_0 = \mathcal{N}(0, I_d)$, whose potential is the quadratic part of U . Because U is separable, the target factorizes into k identical double wells and $2^k - k$ Gaussians, so it has exactly $2^k = d$ modes, and the optimal transport map is diagonal. The source sits exactly on the barrier of every double well, so the untrained pushforward covers none of the 2^k mode combinations. We sweep $k = 1, \dots, 8$, i.e. $d = 2, 4, 8, 16, 32, 64, 128, 256$.

The flow is a neural spline flow on $[-4, 4]^d$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256). Training uses Adam at learning rate 10^{-3} with batch size 200, drawn from a source set of $10^4 \cdot 2^k$ samples; forward KL is sensitive to the batch size, so the small batch sharpens the contrast between the losses. The number of steps decreases with d from 3000 at $d \leq 4$ to 1200 at $d \geq 64$. The μ -surrogate is the same one importance-weighted resampling pass followed by 50 Langevin steps used in the 2D experiments, and $\hat{\mu}$ is built once by QT from $100 \cdot 2^k$ samples. The four losses are those of Section S2.1, with $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$.

Table S4 reports the final ESS of each loss across the eight dimensions. Unlike the 2D mode discovery targets, the flow reaches all 2^k modes for every loss, each mode filled to within a few percent of its uniform 2^{-k} share and the $\kappa = 5$ coverage against $\hat{\mu}$ at least 0.94 throughout, so this is the calibration regime of Annulus, Python, and Chessboard rather than the fake ESS regime, and ESS is the sole discriminator. Three trends emerge. First, the bare forward KL degrades monotonically with dimension, from 0.96 at $d = 2$ to 0.38 at $d = 256$, as the log-weight variance accumulates one double well at a time. Second, the X functionals counteract this and the gap widens with dimension up to $d = 128$: negligible at $d = 2$, it reaches +0.21–0.22 at $d = 64$ and 128, and +0.18 at $d = 256$ where the wide-coverage $X_{\hat{\mu}}$ becomes the best loss. Third, the QT-driven losses, $X_{\hat{\mu}}$ and the mixture, overtake X_{μ} alone at high dimension: X_{μ} tracks the mixture through $d = 64$ but falls below the bare KL by $d = 256$, 0.33 versus 0.38, because it reweights the live one-step IS surrogate whose pushforward is poorly formed early in training, while $X_{\hat{\mu}}$ and the mixture hold their lead because the fixed QT set keeps the flow anchored on the wells even as the log-weight variance grows.

At $d = 256$, i.e. $k = 8$, although the nominal dimension is 256 the *essential* dimension of the target is only $k = 8$: the potential (S14) is a double well in the first 8 coordinates and a standard Gaussian in the remaining 248, so the multimodality lives in an 8-dimensional subspace, while the log-weight variance, and hence the ESS, still loads on all 256. Here we train a single flow and sweep the AIS surrogate ladder length $M \in \{1, 2, 4, 8\}$ that builds the μ -surrogate for the mixture loss, the inner ladder of Algorithm 1, distinct from the temperature ladder of Section 4. Increasing M accelerates convergence in the first training stage, leaving the low ESS plateau sooner, but does not affect the long-time limit: the ESS curves differ only in the first stage and settle at a comparable value (Figure S6). The ladder length is therefore a first-stage accelerator rather than a determinant of the final flow quality.

loss \ d	2	4	8	16	32	64	128	256
forward KL	0.958	0.937	0.935	0.871	0.790	0.669	0.547	0.384
KL+ X_{μ}	0.998	0.983	0.972	0.961	0.893	0.845	0.657	0.333
KL+ X_{μ} + $X_{\hat{\mu}}$	0.987	0.990	0.973	0.955	0.948	0.835	0.750	0.565
KL+ X_{μ} + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.986	0.992	0.975	0.969	0.953	0.878	0.767	0.543

Table S4: Final ESS of each loss on the product multi-well target (S14) across dimension $d = 2^k$, $k = 1, \dots, 8$; the best per column is in bold.

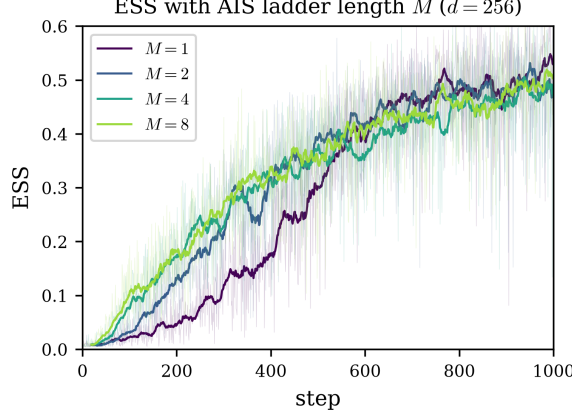


Figure S6: ESS trajectory across training steps for the mixture loss $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$ on the product multi-well at $d = 256$, as the AIS ladder length M that builds the μ -surrogate is varied over $\{1, 2, 4, 8\}$. Faint lines are the raw per-step ESS; bold lines are a moving average.

S2.4 Lattice ϕ^4 field theory: fake ESS at a collective barrier

The scalar ϕ^4 lattice theory is a standard testbed on which flow-based samplers are known to collapse. We use its *tilted* broken phase as a sharp fake ESS instance. The target is the scalar ϕ^4 field on a periodic $L \times L$ lattice, one real value ϕ_j per site, $\phi \in \mathbb{R}^d$ with $d = L^2$, distributed as $\mu \propto e^{-U}$ with action

$$U(\phi) = -2\kappa_{\text{lat}} \sum_{\langle jl \rangle} \phi_j \phi_l + \sum_{j=1}^d \left[\phi_j^2 + \lambda_4 (\phi_j^2 - 1)^2 \right] + h \sum_{j=1}^d \phi_j,$$

with $\langle jl \rangle$ running over nearest-neighbor pairs. In the broken phase the lattice orders into two aligned states near $\pm v \mathbf{1}$, the uniform configurations of field value $\pm v$ ($\mathbf{1}$ the all-ones vector), separated by a *collective* barrier, the free energy of the domain wall any continuous path between them must build; the order parameter is the magnetization $m(\phi) = \frac{1}{d} \sum_{j=1}^d \phi_j$, whose distribution is sharply bimodal. The tilt h breaks the symmetry and carries the design: at $h = 0$ the two vacua hold weight $\frac{1}{2}$ each *by symmetry*, so a collapsed flow loses nothing that symmetrization cannot restore, and equivariant architectures bypass the failure; with $h \neq 0$ the relative weight $p_+/p_- = e^{-\Delta F}$, with $p_+ = \mathbb{P}(m > 0)$ the weight of the minority phase, is a nontrivial number that the sampler must reproduce. We report $L = 6$ and $L = 8$. The parameters are frozen by a classical pilot scan: $\lambda_4 = \frac{1}{2}$ throughout; $\kappa_{\text{lat}} = 0.40$, $h = 0.0257$ at $L = 6$ and $\kappa_{\text{lat}} = 0.40$, $h = 0.0144$ at $L = 8$, placing the measured barrier at 8.0 and $10.2 k_B T$ and the minority-phase weight at $p_+ = 0.128$ and 0.139 . The reference is parallel tempering with the Metropolis-adjusted Langevin algorithm (MALA) over an action-scaling ladder, completing 3.7×10^3 – 4.9×10^4 ladder round trips per run.

The protocol is the single-flow pipeline of Section S2.1, unchanged: a neural spline flow on $[-3, 3]^d$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256); source $\mathcal{N}(0, 0.25 I_d)$ centered on the barrier top; batch size 500, 2000 Adam steps at learning rate 10^{-3} ; the μ -surrogate is the one importance-weighted resampling pass followed by 50 Langevin steps; $\hat{\mu}$ is built once by QT on U , which lands in both vacua from 45–51% of melted starts. Training carries a stability guard: a non-finite loss or gradient skips the gradient step under gradient clipping. Single-flow training is deliberate: a large lattice calls for the adaptive-temperature generator of Section 4, whose hot rungs mix the vacua by construction and would dissolve the failure under study; this benchmark isolates the regime in which the ESS diagnostic itself is misleading.

Table S5 and Figure S7 report the outcome. Bare forward KL and $\text{KL} + \text{X}_\mu$ collapse at every size and every seed: the AIS chain initialized at the pushforward cannot cross an 8.0 – $10.2 k_B T$ collective barrier, so the loss never sees the second vacuum, the reweighted p_+ is exactly zero or exactly one, and the estimated ΔF is infinite where the truth is $+1.8$ – $1.9 k_B T$. The collapse direction is unpredictable across seeds: at $L = 6$ one seed of $\text{KL} + \text{X}_\mu$ lands on the *minority* vacuum at a fake ESS of 0.93, and at $L = 8$ both non-QT-driven losses do so, so ΔF comes out wrong in *sign*. The fake ESS is high and reproducible, 0.81 ± 0.01 over seeds for $\text{KL} + \text{X}_\mu$ at $L = 8$, and arises as follows: X_μ penalizes the spread of z over the batch, and with every

	$L = 6$ ($d = 36$)		$L = 8$ ($d = 64$)	
	ESS	p_+	ESS	p_+
forward KL	0.77 ± 0.04	0, 0, 0	0.57 ± 0.09	0, 0, 1
KL+ X_μ	0.93 ± 0.01	0, 0, 1	0.81 ± 0.01	0, 0, 1
KL+ X_μ + $X_{\hat{\mu}}$	0.88 ± 0.01	0.126 ± 0.001	0.66 ± 0.02	0.127 ± 0.001
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.89 ± 0.01	0.124 ± 0.002	0.68 ± 0.02	0.127 ± 0.001
PT reference	—	0.128	—	0.139

Table S5: Final ESS and reweighted minority-phase weight p_+ on the tilted ϕ^4 lattice, three seeds per size; for the collapsed losses the p_+ column lists the per-seed outcomes, 1 denoting collapse onto the *minority* vacuum. Bold marks the recovered phase weights.

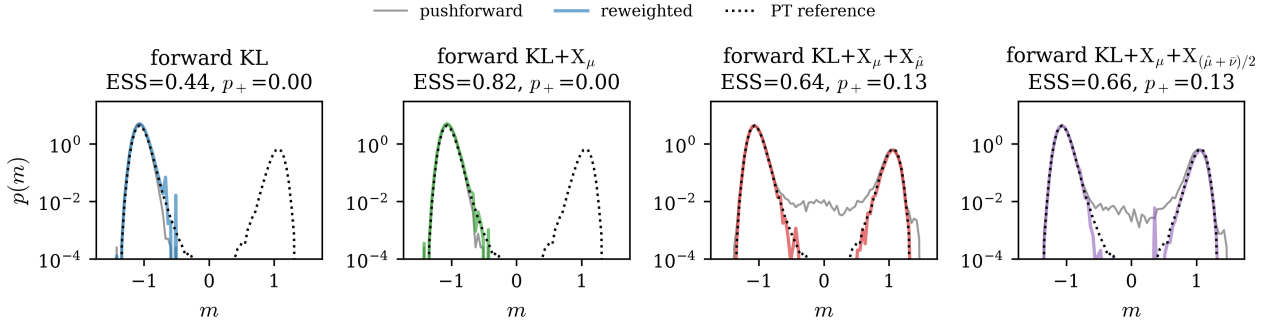


Figure S7: Magnetization density $p(m)$ on the $L = 8$ lattice, one panel per training loss. Each panel shows the pushforward density (gray), the reweighted density (color), and the PT reference (dotted); the title reports the final ESS and the reweighted p_+ .

batch confined to one well it sharpens the in-well fit, reducing the weight variance on the captured support; a variance-reducing term applied to a truncated support raises the restricted ESS while doing nothing for coverage. The ESS is thus highest where the collapse is most severe. The QT-driven losses repair both sizes at every seed: p_+ lands within 0.012 of the reference with seed spread at most ± 0.002 , the minority well is genuinely well sampled rather than rescued by reweighting, with per-mode ESS near or above 0.50 at $L = 8$, and the cost over the collapsed runs is modest, ESS 0.66–0.68 against 0.81. This ESS gap has a topological reading: a diffeomorphism maps the connected Gaussian source to a connected pushforward, visible as the strictly positive gray line across the barrier in Figure S7, so any flow covering both vacua must thread mass through the inter-phase region where the target is exponentially small; that bridge carries near-zero weights, so a flow covering both vacua has a lower ESS than a collapsed one. A training signal driven by ESS alone would not favor covering both vacua; $X_{\hat{\mu}}$ forces the discovery regardless. The repaired weight tracks the action, not the QT set: retraining at $L = 6$ with QT sets deliberately skewed to 5% and 25% minority composition returns $p_+ = 0.124$ – 0.128 throughout, the sample-robustness property of X_ω in action. Stability-guard skips are confined to the QT-driven losses (25% for $X_{\hat{\mu}}$, 5% for the mixture at $L = 8$, zero for both collapsed losses), the early-training cost of confronting cross-well log-ratio spreads.

The same pattern persists at larger lattices but weakens: at $L = 12$ the collapsed losses still discard the minority phase while their ESS falls to 0.15, and the QT-driven losses still cover both phases, recovering p_+ to within 0.02 for $X_{\hat{\mu}}$ and 0.07 for the mixture at ESS 0.01–0.06, correct in kind but expensive and drifting in weight. The fake ESS is therefore most pronounced at moderate size, where it is high enough to mislead; beyond it, single-flow training of either loss family is no longer appropriate and the annealed generator of Section 4 takes over.

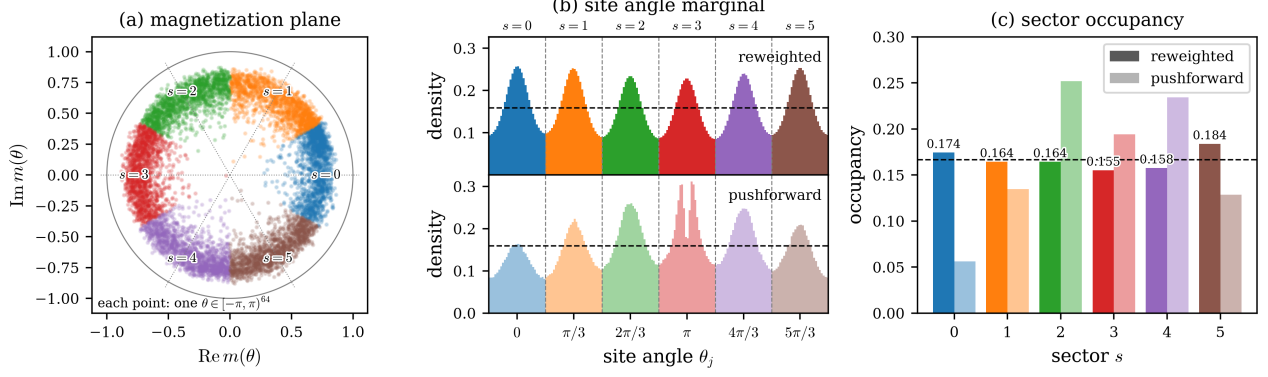


Figure S8: Structure of the cold clock measure (S15) at $d = 64$, displayed on 10^6 reweighted samples: fresh source draws advanced through the trained ladder by the map-reweight-resample step (v), X-regularized loss at $B = 10^4$. (a) Global magnetization $m(\theta)$, colored by sector index $s(\theta)$, 1500 points per sector: the measure splits into $p = 6$ equivalent sectors with mean $|m| = 0.73$. (b) Site-angle marginal: the anisotropy locks each site to the six clock angles $2\pi s/6$, $s = 0, \dots, 5$; dashed lines bound the six wells, colored as in (c), and the black line marks the uniform density $\frac{1}{2\pi}$; the pushforward marginal is from the maps alone, without reweighting. (c) Sector occupancy of reweighted and pushforward samples, the black line marking the uniform occupancy $\frac{1}{6}$.

S2.5 Boltzmann generator on p -state clock model

The preceding experiments train a single flow map; this experiment exercises the full adaptive-temperature procedure of Algorithm 4, contrasting the X-regularized stage loss (16) against the bare forward KL through an otherwise identical pipeline. The target is the p -state clock model on a periodic 8×8 square lattice, one angle per site,

$$U(\theta) = -J \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) - h \sum_{j=1}^d \cos(p\theta_j), \quad \theta \in [-\pi, \pi]^d, \quad d = 64, \quad (\text{S15})$$

with nearest-neighbor coupling $J = 1$, $\mathbb{Z}_p$ anisotropy $h = \frac{1}{2}$, and $p = 6$ clock states; for $p \geq 5$ the model hosts two Berezinskii–Kosterlitz–Thouless (BKT) transitions enclosing a critical quasi-long-range-ordered phase (José et al., 1977). The cold measure $\propto e^{-U}$ splits into $p = 6$ equivalent sectors labeled by the phase of the global magnetization

$$m(\theta) = \frac{1}{d} \sum_{j=1}^d e^{i\theta_j}, \quad s(\theta) = \left\lfloor \frac{p \arg m(\theta)}{2\pi} \right\rfloor \bmod p \in \{0, \dots, 5\},$$

where $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer. The annealing path from the uniform measure to μ traverses the BKT window, where the correlation length grows rapidly and the six sectors deepen simultaneously. Figure S8 displays this structure on the trained validation set.

The source is the uniform measure on the torus, $U_0 = 0$, so $U_t = tU$; each stage map G_k is a circular neural spline flow on $[-\pi, \pi)^{64}$ with 6 coupling transforms, 16 bins, and hidden widths (256, 256). The protocol is deliberately large-sample: a validation set of 10^6 samples, pool size $P = 2 \times 10^5$, safe start $t_{\text{safe}} = 0.2$, shared ladder length $M = 4$, and SMC floor $\tau = 0.7$, which trims over-aggressive guesses before any training is spent. We report three batch sizes $B = 10^3, 10^4, 10^5$, with 2000, 500, 200 Adam steps per stage; the remaining constants are those of Algorithm 4. Training carries a stability guard: a non-finite loss skips the gradient step under gradient clipping and a fresh batch is drawn, with a stage abort only after repeated skips. The two losses compared are the full stage loss (16) at the balanced hyperparameters $\lambda = 1$, $\alpha = \beta = \frac{1}{2}$, and the bare forward KL, its first term alone; everything else is shared.

Table S6 reports the core metric, the validation ESS of step (v) on every accepted rung. Both losses complete the ladder at every batch size, along *identical* ladders with *identical* rejected rungs: the adaptive

machinery converts loss quality into schedule, not into success or failure. Figure S9 shows the per-step training ESS on the first three stages. The discrimination lives in the per-rung quality. On accepted rungs the X-regularized loss posts the higher validation ESS on stages 1–3 at all three batch sizes, +0.21/+0.13/+0.10 at $B = 10^3$, +0.17/+0.11/+0.03 at $B = 10^4$, +0.13/+0.09/+0.04 at $B = 10^5$; the gap closes to zero in the transition window, slightly reversing on a few late rungs. Wall time per pair is equal within 3%, X-regularized versus bare: 51.9 versus 50.7 minutes at $B = 10^3$, 35.7 versus 34.8 at $B = 10^4$, and 125.5 versus 122.8 at $B = 10^5$; every attempted rung trains the same number of Adam steps, so a rung costs the run average of 3.4, 4.4, and 15.5 minutes respectively, over 15, 8, and 8 attempted rungs. A generalization check pushes 10^6 independent source samples through the trained $B = 10^4$ ladders, stage by stage with the resampling of step (v): the fresh-sample per-stage ESS tracks the accepted-rung validation ESS of Table S6 within 0.02–0.10 for both losses, and the final sector occupancy is uniform to ± 0.02 per sector, total variation 0.025, so the gates measure generalization rather than memorization of the carried set.

stage k	1	2	3	4	5	6	7	8
t_k ($B = 10^3$)	0.200	0.396	0.530	0.662	0.753	0.841	0.952	1.000
forward KL	0.391	0.398	0.463	0.366	0.467	0.465	0.400	0.708
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.602	0.524	0.559	0.401	0.500	0.453	0.369	0.778
t_k ($B = 10^4$)	0.200	0.396	0.588	0.720	0.849	1.000	—	—
forward KL	0.531	0.523	0.347	0.396	0.349	0.341	—	—
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.703	0.628	0.381	0.412	0.336	0.303	—	—
t_k ($B = 10^5$)	0.200	0.396	0.588	0.720	0.849	1.000	—	—
forward KL	0.611	0.602	0.416	0.461	0.400	0.387	—	—
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$	0.745	0.688	0.451	0.489	0.405	0.383	—	—

Table S6: Validation ESS on every accepted rung, the acceptance gate of Algorithm 4 with floor $\tau_v = 0.3$, on the $p = 6$ clock model at $d = 64$; bold marks the better loss on each rung. Within each pair the two losses traverse the same accepted ladder and reject the same rungs, seven rejections per run at $B = 10^3$ and two at $B = 10^4$ and 10^5 .

The pattern is consistent with the theory: the X terms lower the asymptotic accuracy floor of each stage (Section S1.3), a larger training *limit*, but promise no faster transient, and the rejected-rung values confirm that the bare forward KL trains no slower. The anatomy of the ladder shows where the gain lives. The early bridges, $t \leq 0.66$, are far but smooth: a large, structurally simple deformation with a clean AIS signal, so the training loss is the binding constraint and the X terms win by up to +0.21. In the transition window, $t \approx 0.66$ –0.95, the selector halves the accepted increment, an empirical readout of the annealing bottleneck at the BKT crossing; there the ceiling is set by the critically slow Langevin rejuvenation and the shared network capacity, constraints common to both losses, so the gap collapses to zero. The final hop to $t = 1$ is the easiest rung for both. On this target the six sectors are never isolated (Figure S8(a)), so the mode discovery channel of $X_{\hat{\mu}}$ is not load-bearing; the qualitative separations of Sections S2.1–S2.3, where modes are isolated by construction, do not transfer to soft multimodality. What transfers is a net per-rung quality improvement, concentrated on the early bridges, at zero extra cost: the X-regularized loss was never worse than the bare forward KL where it mattered and never failed where the bare loss passed.

The trained ladder is used as a staged sampler: each stage applies its map G_k^{-1} and then reweights and resamples as in step (v), and Figure S8(c) shows the resulting occupancy, uniform to ± 0.02 . The residual is nonetheless well above the i.i.d. multinomial floor of 3×10^{-4} at 10^6 samples. The weights of step (v) are pointwise density ratios, exchangeable within a sector and hence sector-blind: they neither detect the imbalance, seeded by the slight flow bias that the i.i.d. pushforward exposes, nor correct it; the cold Langevin rejuvenation transports no mass between sectors; and the final validation set is resampled, carrying no weights at all. The occupancy is thus unbiased in expectation over runs but unrestored within any single one, and the per-stage fluctuations accumulate along the ladder. A scaling test quantifies the accumulation: rebuilding fresh sample sets of size $N = 10^4 \cdot 2^k$, $k = 0, \dots, 7$, through the trained $B = 10^4$ ladder, with

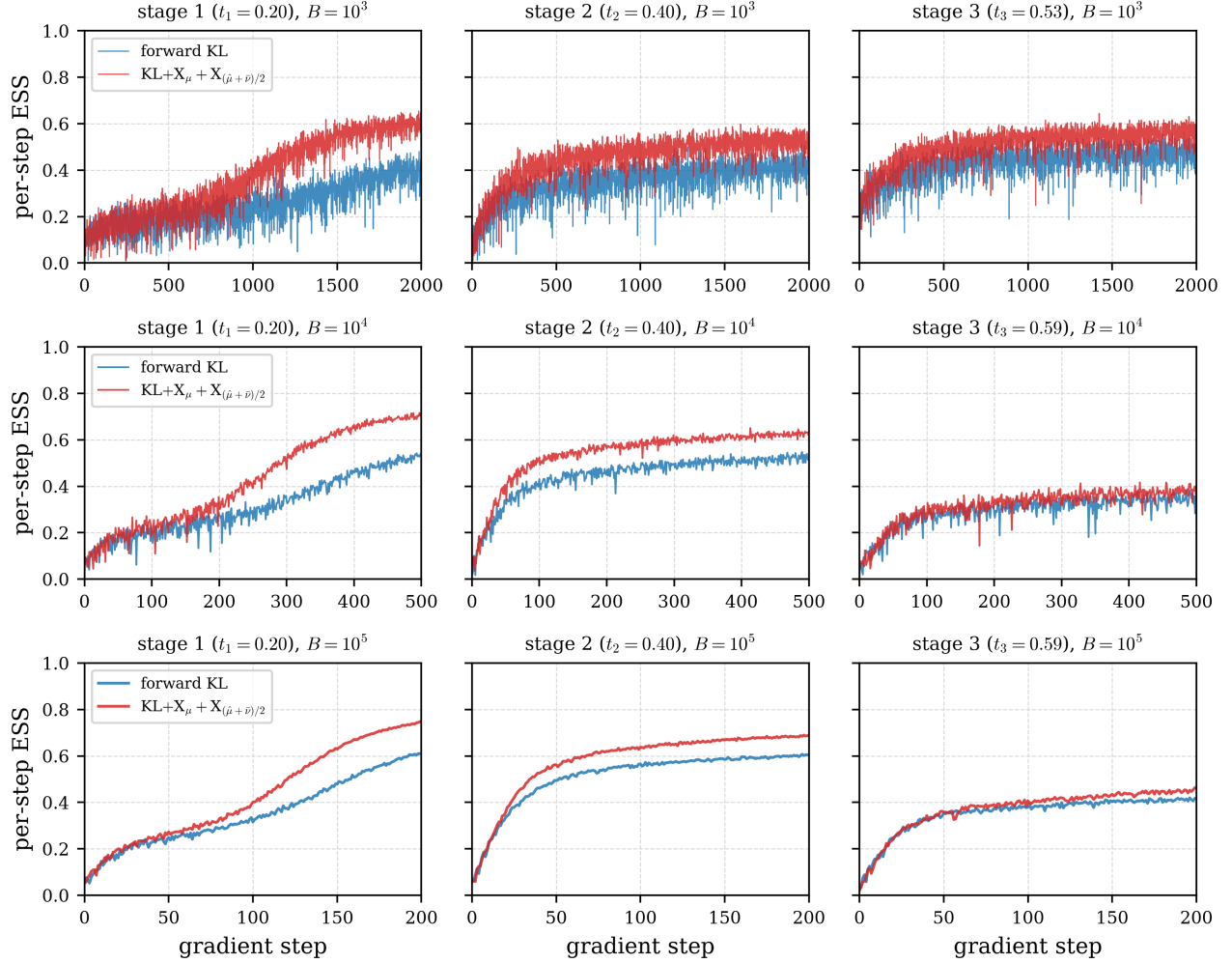


Figure S9: Per-step ESS of the training surrogate on the first three stages, bare forward KL (blue) versus the X-regularized loss (red); rows $B = 10^3, 10^4, 10^5$, one panel per stage.

2^{7-k} independent rebuilds per size, the mean occupancy bias $\frac{1}{6} \sum_s |p_s - \frac{1}{6}|$ contracts from 0.075 at $N = 10^4$ to 0.017 at $N = 1.28 \times 10^6$ at the fitted rate $N^{-0.34}$, shown in Figure S10, distinctly slower than the Monte Carlo rate $N^{-1/2}$ of a single multinomial draw: the staged sampler composes six map–reweight–resample steps, and each resampling correlates the surviving lineages, so the effective number of independent sector draws grows sublinearly in N . Coverage, ESS, and occupancy evenness are three distinct axes of sample quality; the failure that no post-processing could repair, a *missing* sector, does not occur for either loss.

S2.6 Bayesian screened Poisson source inversion

The final experiment applies the adaptive-temperature generator to a Bayesian PDE inverse problem, the hardest target in our suite: a multimodal posterior with an enumerable well structure that the ladder must reproduce with the correct relative weights. The unknown is a periodic field on $[0, 1]^2$,

$$v(\mathbf{x}; \boldsymbol{\theta}) = \sum_{\mathbf{k}} \theta_{\mathbf{k}} \psi_{\mathbf{k}}(\mathbf{x}), \quad \mathbf{x} = (x_1, x_2) \in [0, 1]^2,$$

where each basis function is attached to a wave vector $\mathbf{k} = (k_1, k_2) \in \mathbb{Z}^2$: the constant $\psi_{(0,0)}(\mathbf{x}) \equiv 1$ for $\mathbf{k} = (0, 0)$, and the pair

$$\psi_{\mathbf{k}}(\mathbf{x}) = \sqrt{2} \cos(2\pi(k_1 x_1 + k_2 x_2)), \quad \psi_{\mathbf{k}}(\mathbf{x}) = \sqrt{2} \sin(2\pi(k_1 x_1 + k_2 x_2))$$

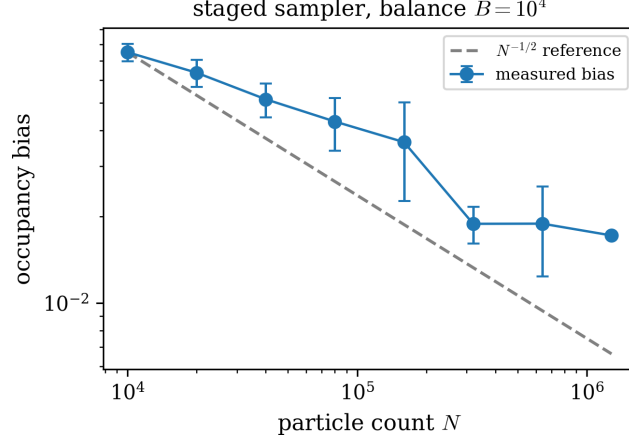


Figure S10: Occupancy bias of fresh staged rebuilds through the trained $B = 10^4$ ladder of the X-regularized loss, versus rebuild size $N = 10^4 \cdot 2^k$, $k = 0, \dots, 7$, averaged over 2^{7-k} independent rebuilds per size at equal total work; error bars are ± 2 standard errors. The dashed line is the Monte Carlo reference $N^{-1/2}$.

for every other $\mathbf{k}$, taken once per $\pm \mathbf{k}$ pair over the half-plane $k_1 > 0$ or ($k_1 = 0, k_2 > 0$); together these form an orthonormal basis of $L^2([0, 1]^2)$. We keep the $d = 36$ Fourier basis functions of smallest $k_1^2 + k_2^2$, an isotropic cutoff. The prior is Gaussian with a steeply decaying spectrum, $\theta_{\mathbf{k}} \sim \mathcal{N}(0, \sigma_{\mathbf{k}}^2)$ with $\sigma_{\mathbf{k}}^2 = 2/(1 + |\mathbf{k}|^2)^6$, so this truncation already captures essentially all of the prior variance. The field drives a screened Poisson equation with a periodic cosine source,

$$(-\Delta + 25)u = 3\left(1 + \frac{1}{2}\cos(3v(\cdot; \theta))\right), \quad (\text{S16})$$

solved spectrally on a 32×32 grid, and the observation operator $\mathcal{F}(\theta) = (u(\mathbf{x}_j; \theta))_{j=1}^{54}$ reads the solution $u(\cdot; \theta)$ of (S16) at 54 sensors on the ring of radius 0.3 about the center, more sensors than unknowns; Figure S11(a,b) shows the field $v(\mathbf{x}; \theta^\dagger)$ and the observed solution. The observation geometry drives much of the difficulty: all sensors lie on a single small circle, so distinct source configurations produce nearly identical data on the ring, and the posterior concentrates on the resulting family of near-equivalent wells. The data are $\mathbf{y} = \mathcal{F}(\theta^\dagger) + \sigma_{\text{obs}} \boldsymbol{\eta}$ with $\theta^\dagger$ a fixed prior draw and $\boldsymbol{\eta}$ standard normal.

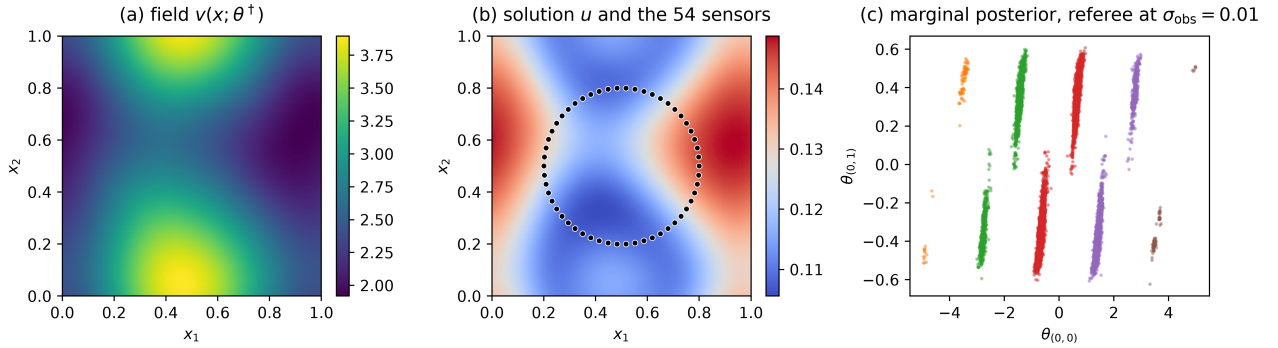


Figure S11: The screened Poisson inverse problem. (a) The field $v(\mathbf{x}; \theta^\dagger)$ on $[0, 1]^2$, whose cosine modulation $3(1 + \frac{1}{2}\cos(3v(\mathbf{x}; \theta^\dagger)))$ is the source. (b) The corresponding solution u of (S16) and the 54 sensors on the ring of radius 0.3, the dotted circle, where u is observed. (c) The marginal posterior in the coefficients $(\theta_{(0,0)}, \theta_{(0,1)})$ from the parallel tempering referee at $\sigma_{\text{obs}} = 0.01$, colored by the shift index n : the posterior splits into the well lattice of the cosine symmetries.

All sampling runs on whitened coordinates $\xi_{\mathbf{k}} = \theta_{\mathbf{k}}/\sigma_{\mathbf{k}}$, on which the posterior potential reads

$$U(\xi) = \frac{1}{2}|\xi|^2 + \Phi(\sigma \odot \xi), \quad \Phi(\theta) = \frac{|\mathbf{y} - \mathcal{F}(\theta)|^2}{2\sigma_{\text{obs}}^2},$$

with $\sigma = (\sigma_k)$ the vector of prior standard deviations and $\odot$ the element-wise product. The ladder runs with the d -dimensional whitened prior as its source, $U_0(\xi) = \frac{1}{2}|\xi|^2$, so the Algorithm 4 bridge tempers exactly the likelihood, $(1-t)U_0 + tU = \frac{1}{2}|\xi|^2 + t\Phi$. The cosine makes the posterior multimodal with an enumerable structure. Writing $\theta_{(0,0)} = \int_{[0,1]^2} v(\mathbf{x}; \theta) d\mathbf{x}$ for the coefficient of the constant basis function $\psi_{(0,0)} \equiv 1$, the spatial mean of the field, a shift $v \rightarrow v + 2\pi n/3$ acts on this mode alone, $\theta_{(0,0)} \rightarrow \theta_{(0,0)} + 2\pi n/3$. Since $\cos(3v)$ is invariant under $v \rightarrow -v$ and under these shifts, the wells form a lattice indexed by the shift integer $n = \text{round}(3\theta_{(0,0)}/2\pi)$ and a sign, around ten of which carry visible mass, shown in Figure S11(c). The prior tilts the shift weights and the truth draw tilts the signs, so the well weights are nontrivial numbers. A parallel tempering MALA referee, 20 temper rungs and 128 chains certified by 7 300–35 000 ladder round trips per noise level, supplies the reference weights.

The ladder runs in dimension $d = 36$; each stage map acts on all 36 whitened coordinates. Every stage pushes its validation samples through the stage inverse and scores them by the stage importance weight $\log w_k(\xi) = U_{t_{k-1}}(G_k(\xi)) - U_{t_k}(\xi) - \log |\det J_{G_k}(\xi)|$ of step (v); the ESS of w is the per-stage validation ESS on which the acceptance gate of Algorithm 4 fires. The protocol is otherwise Algorithm 4 with NSF stage flows on $[-5, 5]^{36}$, batch size 4 000, 1 000 Adam steps per stage, $M = 6$ surrogate rungs, selection floor $\tau = 0.5$, validation floor $\tau_v = 0.3$, and at most ten attempts per stage. The observation noise is the difficulty dial: shrinking σ_{obs} sharpens the likelihood by $1/\sigma_{\text{obs}}^2$, deepens the barriers, and widens the annealing gap, leaving the well lattice and the referee intact. We sweep $\sigma_{\text{obs}} \in \{0.01, 0.015, 0.02, 0.025\}$ and run the bare forward KL and the X-regularized stage loss (16), $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ at balanced hyperparameters, on each.

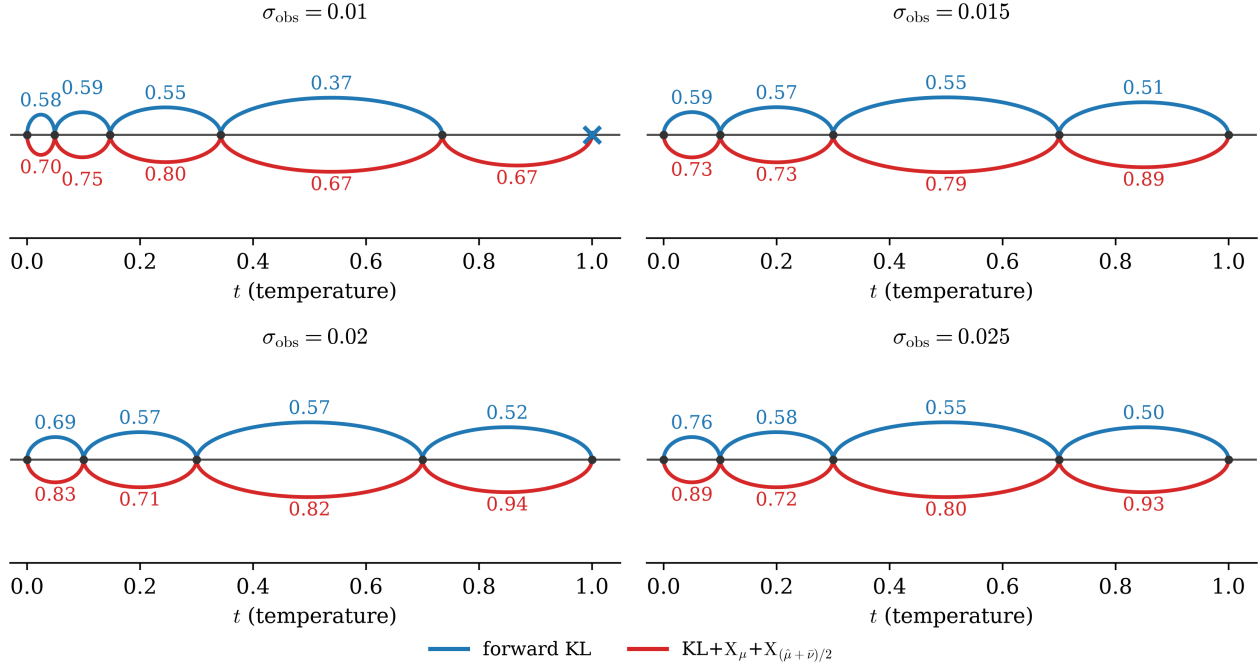


Figure S12: ESS ladders on the screened Poisson posterior across the noise sweep, one panel per σ_{obs} . The temperature axis runs $t = 0 \rightarrow 1$; each accepted stage is an arc spanning its increment Δt_k , forward KL above (blue) and the X-regularized loss $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ below (red), labelled by its per-stage validation ESS. The cross marks the final bridge that the forward KL run failed in all ten attempts at $\sigma_{\text{obs}} = 0.01$. The recovered well weights are reported in Table S7.

Figure S12 reports the outcome in three layers. First, existence: at the sharpest noise the bare forward KL cannot finish the ladder; its per-stage ESS starts at 0.58 and falls to 0.37 on the final accepted rung, and the bridge to $t = 1$ fails all ten retraining attempts, while the X-regularized loss completes the same rungs at 0.67–0.80 in 39 minutes against 95 spent failing. Second, per-stage quality: at every noise level the X-regularized loss leads on every rung; on the final rung of the three completed sweeps the X-regularized loss reaches 0.89–0.94 against the forward KL’s 0.50–0.52. The separation is considerably larger than on the p -state clock model of Section S2.5, where both losses completed identical ladders and the X terms improved

σ_{obs}	forward KL	KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$
0.010	0.106 ± 0.039	0.042 ± 0.002
0.015	0.077 ± 0.016	0.041 ± 0.004
0.020	0.028 ± 0.010	0.020 ± 0.001
0.025	0.050 ± 0.004	0.019 ± 0.003

Table S7: Total-variation distance of the staged sampler’s well weights from the parallel tempering reference, across the noise sweep; mean and spread over three independent replays of the trained ladder. Bold marks the smaller distance of each pair. The forward KL run at $\sigma_{\text{obs}} = 0.01$ did not reach $t = 1$ (Figure S12); its weights are read from the last accepted stage.

a rung by at most +0.21: the PDE posterior is closer to ill-posedness, its prior-to-posterior gap growing as $1/\sigma_{\text{obs}}^2$, and on such ladders the training signal rather than the sampling physics is the binding constraint, and the X terms are designed to strengthen the training signal. Third, weight accuracy (Table S7), measured by replaying the trained ladder as the staged sampler of step (v) three times per run: with the gates passed and the ladders complete, the bare forward KL produces well weights at total variation 0.028–0.077 from the certified reference, with a replay spread up to ± 0.016 ; the X-regularized loss reproduces the reference at 0.019–0.042 with replay spread within ± 0.004 , closer and roughly $4\times$ more reproducible, with up to $2.6\times$ smaller total-variation error where the forward KL completes the ladder. A completed ladder with per-stage ESS above the validation floor does not guarantee correct well weights: the per-stage diagnostic is local in temperature and blind to the slow drift of the carried set, an accumulation of the same kind as the occupancy drift of Section S2.5, though here it lives in the relative well weights, and only the QT-driven mixture keeps them pinned. The error originates in the carried validation set: each stage resamples it by the trained flow’s weights, so a small per-stage bias in the relative well masses propagates into every later stage, and the per-stage ESS does not register it. Both losses cover all ten wells that the referee certifies above the 10^{-3} level in every run; the carried validation set preserves the support, and the failure is confined to the density fit and the weights, the regime the X_μ and mixture terms address.

S3 Boltzmann generators on molecular potential energies

The model problems of the previous section were each chosen to isolate one failure mode of forward KL training on a smooth target: the 2D benchmarks, the lattice ϕ^4 theory, and the screened Poisson inversion. The application that motivates a Boltzmann generator in the first place is a molecule, and a molecular target exercises the whole pipeline at once. This section applies the adaptive-temperature, X-regularized generator of the previous section to three molecules of increasing difficulty in full internal coordinates, $d = 36$ to 60, on a single consumer GPU. It is organized into five parts: the motivation and the two difficulties that make a molecular potential hard (Section S3.1); *sharpening*, the core simplification that makes the framework tractable on a singular target (Section S3.2); the molecule-specific engineering it requires (Section S3.3); the numerical results on glycerol, diethanolamine, and alanine dipeptide (Section S3.4); and the limitations of the molecular study (Section S3.5).

S3.1 Motivation and background

The distribution that motivates a Boltzmann generator is a molecule: the equilibrium (Boltzmann) distribution $\mu \propto e^{-U}$ of a molecular potential, with U its dimensionless reduced form, whose samples are the conformers that set binding affinities, reaction rates, and free energies in computational chemistry and drug design (Noé et al., 2019). The targets studied so far were smooth by design; a molecular potential is not. Drawing *independent* samples from μ is the central computational problem: MD and MCMC mix slowly across the high torsion barriers that separate conformers, so a trained flow that emits independent conformers with a tractable density, and therefore supports the exact importance-sampling pipeline of the previous sections, is the payoff that justifies the training cost.

We take the potential energy to be the Amber molecular-mechanics force field (Cornell et al., 1995), the standard classical model, with raw force-field energy $U_{\text{FF}}(x)$ in kJ/mol, a sum of bonded terms, harmonic

bonds and angles and periodic torsions, and nonbonded Lennard-Jones and Coulomb terms,

$$U_{\text{FF}}(x) = \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_\theta(\theta - \theta_0)^2 + \sum_{\text{torsions}} \frac{V_n}{2} [1 + \cos(n\phi - \gamma)] + \sum_{i < j} \left[4\epsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^6}{r_{ij}^6} \right) + \frac{q_i q_j}{r_{ij}} \right], \quad (\text{S17})$$

the last sum running over nonbonded atom pairs, with charges in units that absorb the Coulomb constant. The Boltzmann distribution at temperature T is $\mu \propto e^{-U}$ in the dimensionless reduced potential

$$U(x) = \frac{1}{k_B T} U_{\text{FF}}(x),$$

with k_B the Boltzmann constant. We evaluate U_{FF} for the *isolated molecule in vacuum*, with no solvent and no nonbonded cutoff; this and two further simplifications of the molecular study are stressed in Section S3.5.

We work in *whitened internal coordinates* (Noé et al., 2019). The six rigid-body degrees of freedom, global translation and rotation, leave U unchanged, so we drop them and describe the shape by bond lengths, bond angles, and torsions, a z-matrix giving a smooth, invertible map $x \mapsto \xi \in \mathbb{R}^d$ with $d = 3N - 6$ for an N -atom molecule and a tractable volume Jacobian. Bonds and angles are stiff and near-Gaussian about their equilibria, so each is standardized to unit variance; torsions stay periodic on $[-\pi, \pi)$. The source $\mu_0 \propto e^{-U_0}$ is then a trivial product base, standard normal on the bond/angle directions and uniform on the torsions, and the target, pulled back through the change of variables, is

$$\mu(\xi) \propto \exp(-U(x(\xi))) \left| \det(\partial x / \partial \xi) \right|,$$

so U_0 and U as functions of ξ already absorb the change-of-variables log-Jacobian. This is what makes the target tractable for a flow: the rigid-body symmetries and the stiff, strongly correlated bond/angle vibrations are factored out, leaving the combinatorially multimodal torsional landscape, the conformers, as the structure the diffeomorphism must actually transport.

Two features make (S17) qualitatively harder than a smooth model target, and they compound. First, the dimension grows with the molecule ($d = 36$ to 60 here), and the conformer landscape is *combinatorially multimodal*: every rotatable torsion is multi-well (gauche $^\pm$ /trans), so the joint over several torsions has dozens of basins the flow must discover. Second, and more consequential, the Lennard-Jones r^{-12} repulsion makes U *singular*: as any pair distance $r_{ij} \rightarrow 0$ the energy diverges, so the target has a hard, near-singular clash boundary that a smooth diffeomorphism cannot represent without enormous Jacobians. Neither difficulty is decisive in isolation, since high-dimensional smooth targets and low-dimensional singular ones are both routine, but together they are worse than either alone: the singular clash core sits *inside* the high-dimensional multimodal landscape, and a single flow asked to discover modes *and* resolve the singularity at once trains badly at both. This compounding of the two difficulties, absent from every model problem of the previous section, is what the rest of this section addresses. On alanine dipeptide ($d = 60$) below it is decisive: the clash core, not the torsion profile, becomes the binding constraint.

S3.2 Sharpening: the core simplification

Sharpening keeps the singular part out of the flow entirely, by annealing a second index the flow never sees. Annealing runs from the source U_0 of Section S3.1 to the target, for which we now write the alias $U_1 = U$. A *regularization* index $s \in [0, 1]$ defines a soft-core target $U_{1,s}$ by taming the $r \rightarrow 0$ clash of the raw energy U_{FF} (S17) before its reduction by $k_B T$. Two crucial soft-core parameters act on U_{FF} : an energy cap e_{cap} (kJ/mol) that log-compresses the energy above its threshold to $e_{\text{cap}} + a \log(1 + (U_{\text{FF}} - e_{\text{cap}})/a)$ with fixed scale $a = 50$ kJ/mol, a C^1 wall of finite height and gradient (a hard clamp would zero the escape force); and a distance floor r_{floor} (nm) that replaces each nonbonded pair distance by $\max(r_{ij}, r_{\text{floor}})$, removing the r_{ij}^{-12} pole. Writing $U_{\text{FF}}|_{e_{\text{cap}}, r_{\text{floor}}}$ for the resulting regularized energy, the reduced soft-core target is

$$U_{1,s} = \frac{1}{k_B T} U_{\text{FF}}|_{e_{\text{cap}}(s), r_{\text{floor}}(s)}, \quad (\text{S18})$$

with the cap threshold $e_{\text{cap}}(s)$ and floor $r_{\text{floor}}(s)$ annealed from soft ($s = 0$) to sharp ($s = 1$) on a geometric or arithmetic schedule,

$$e_{\text{cap}}(s) = e_{\text{min}}(e_{\text{max}}/e_{\text{min}})^s \quad \text{or} \quad e_{\text{min}} + (e_{\text{max}} - e_{\text{min}})s, \quad r_{\text{floor}}(s) = r_{\text{max}} + (r_{\text{min}} - r_{\text{max}})s,$$

between a low cap $e_{\min}$ and high cap $e_{\max}$, and a loose floor $r_{\max}$ and the physical floor $r_{\min}$. Both controls act only on the high-energy clash region, inactive on the equilibrium basins where $U_{\text{FF}} \leq e_{\text{cap}}$ and $r_{ij} > r_{\text{floor}}$, so $U_{1,1} = U_1$ there and $U_{0,s} = U_0$ for all s . The bridge potential carries both indices,

$$U_{t,s} = (1-t)U_0 + tU_{1,s},$$

and the whole run is a single anneal from the soft source $U_{0,0}$ to the sharp target $U_{1,1}$, the regularization reaching the temperature at every accepted stage, $s_k = t_k$. The two indices coincide only at the accepted nodes: within a stage they differ, t first advancing to t_k while s lags at s_{k-1} , then s catching up to s_k , as described next.

The change from Algorithm 4 is one extra, flow-free step per stage. Algorithm 4 advances the single temperature index t on the bridge U_{t_k} ; with sharpening each stage advances the pair (t, s) in two sub-steps,

$$\underbrace{(t_{k-1}, s_{k-1}) \xrightarrow{G_k} (t_k, s_{k-1})}_{\text{flow step: advance } t} \xrightarrow{w_k^{\text{sharp}}} \underbrace{(t_k, s_{k-1}) \rightarrow (t_k, s_k)}_{\text{sharpening step: advance } s}.$$

The *flow step* advances the temperature at the *lagged* regularization, $U_{t_{k-1}, s_{k-1}} \rightarrow U_{t_k, s_{k-1}}$: at fixed s_{k-1} the soft-core bridge $U_{t, s_{k-1}}$ is a single-temperature bridge to the target $U_{1, s_{k-1}}$, so the trained map G_k and its validation update are exactly steps (i)–(v) of Algorithm 4 with U_1 replaced by $U_{1, s_{k-1}}$; the diffeomorphism only ever transports the smooth surrogate. The *Monte Carlo sharpening step* then advances the regularization at fixed temperature, $U_{t_k, s_{k-1}} \rightarrow U_{t_k, s_k}$, by a single importance reweight with weight

$$w_k^{\text{sharp}}(\xi) = \exp(U_{t_k, s_{k-1}}(\xi) - U_{t_k, s_k}(\xi)) = \exp(-t_k[U_{1, s_k}(\xi) - U_{1, s_{k-1}}(\xi)]), \quad (\text{S19})$$

followed by resampling and a short Langevin rejuvenation on U_{t_k, s_k} , with no flow. The importance step asks nothing the singularity forbids: it never inverts or differentiates U_1 , only reweights and rejuvenates samples inside the basins they already occupy, so the stiff repulsive wall costs a little weight variance rather than the unbounded Jacobian it would impose on a smooth map. This is the asymmetry the split exploits: the flow handles the part that is smooth but high-dimensional, the Monte Carlo step the singular part, confined to the thin clash boundary.

Each stage thus yields two effective sample sizes: the flow’s validation $\text{ESS}_k^{\text{val}}$ from the flow step, which is the ESS of the step-(v) validation weights w_k of Algorithm 4, and the sharpening $\text{ESS}_k^{\text{sharp}}$ of w_k^{sharp} (S19); each is the normalized effective sample size of Section 3.4, a fraction in $(0, 1]$. The Monte Carlo error-propagation factor through the two per-stage reweights,

$$F = \prod_k \frac{1}{\text{ESS}_k^{\text{val}}} \frac{1}{\text{ESS}_k^{\text{sharp}}}, \quad F = 1 \text{ ideal}, \quad (\text{S20})$$

charges the cost of both; it is smaller-is-better, and a run with no sharpening step has $\text{ESS}_k^{\text{sharp}} \equiv 1$. On the clash-dominated target a few near-singular weights collapse the raw validation ESS, so before forming it we saturate the log-weights about their median. With m the median of $\{\log w_i\}$ and $\sigma_w = c \text{median}_i |\log w_i - m|$, the *smooth* validation effective sample size is

$$\text{ESS}_k^{\text{val}*} = \text{ESS}(\{\tilde{w}_i\}), \quad \log \tilde{w}_i = m + \sigma_w \tanh((\log w_i - m)/\sigma_w), \quad (\text{S21})$$

the normalized ESS of Section 3.4 evaluated on the saturated weights, with $c = 6$ on alanine dipeptide as detailed in Section S3.3. Its error-propagation factor is

$$F^* = \prod_k \frac{1}{\text{ESS}_k^{\text{val}*}} \frac{1}{\text{ESS}_k^{\text{sharp}}}. \quad (\text{S22})$$

We report F where the raw validation ESS is well behaved, and F^* , written with a star, where the clash singularity forces the smooth estimator; the two are the same product evaluated with raw or smooth validation ESS. This factor is the headline metric for every result below: the variance amplification a downstream importance-sampling estimator inherits from the full ladder.

S3.3 Engineering for the singularity

The singular targets require three further devices, all inert on the smooth ones (glycerol and diethanolamine ran with none of them). (i) *Particle drop*: each validation and resample ESS discards the worst 2% of particles before the ESS is formed, so a handful of clash-singular outliers, finite but enormous under the soft cap, cannot hijack the importance weights. (ii) *Best ESS gate snapshots*: within a stage the flow is snapshotted at each ESS gate and the stage keeps the *highest* validation ESS snapshot rather than the last training step (which overfits the soft cap and sheds ESS); a step-0 identity snapshot is always included as a Monte Carlo fallback. (iii) *Distance-floor anneal*: for the worst clash core (alanine dipeptide) r_{floor} is annealed alongside the cap, softening the Lennard-Jones wall early and tightening it to the physical value only at $s = 1$, at the cost of a longer ladder. The wide-coverage pool $\hat{\mu}$ is supplied by the δ -reweighted quench and temper of Section 3.3 ($\delta = 0.1$), unchanged; the sharpening split remains the core mechanism, and these three devices are the supporting engineering the singularity demands.

A final, metric-level choice accompanies them on the singular target: the per-stage validation ESS that gates and scores each stage is the smooth estimator $\text{ESS}_k^{\text{val}*}$ of (S21), with $c = 6$ on alanine dipeptide. Saturating the log-weights about their median bounds the few clash-singular dominators that otherwise collapse the raw weighted ESS; a raw ESS of 0.4 then reads as a smooth ≈ 0.5 . Glycerol and diethanolamine, free of such dominators, keep the raw ESS and report F .

S3.4 Numerical results

We close three molecules of increasing difficulty in full internal coordinates on a single consumer GPU (RTX 5070 Ti): glycerol ($d = 36$), diethanolamine ($d = 48$), and alanine dipeptide ($d = 60$, the canonical Boltzmann generator benchmark). Table S8 summarizes the comparison that runs through all three: the X-regularized δ -reweighted loss against an annealed SMC reference, the same adaptive sharpening scheme with the flow set to the identity at every stage, which selects its own, longer ladder. Because the factor is a product over stages, the *ratio* of trained-flow to annealed SMC F , not the absolute value, is the cross-molecule comparison: the trained flow cuts F by $\sim 30\text{--}40\times$ on the two clash-mild, combinatorially multimodal targets, but only $\sim 2\times$ on alanine dipeptide, where the singular clash core rather than the multimodality dominates — the core that the Monte Carlo sharpening already resolves. The per-molecule subsections below give the full ablation, ladder, and reference figures for each. The baselines here, the bare forward KL and the identity-flow annealed SMC, are the natural same-cost comparisons for a forward KL-trained generator.

target	d	K	F/F^*	
			KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$ flow	annealed SMC
glycerol	36	4	14.16	463.02
diethanolamine	48	5	35.90	1455.44
alanine dipeptide	60	11	67.95*	129.70*

Table S8: Molecular Boltzmann generators in full internal coordinates. K is the trained-flow ladder length; the annealed SMC reference adaptively chooses its own, longer ladder, $K = 10, 10, 12$ respectively. Both columns run the same adaptive sharpening scheme, annealing the regularization from the soft source to the sharp target, and differ in whether each stage trains a flow or sets it to the identity. Entries are the error-propagation factor F (S20), or the smooth ESS variant F^* (S22) where starred, used on alanine dipeptide; both include the Monte Carlo sharpening reweight, smaller is better, best in bold.

S3.4.1 Glycerol ($d = 36$)

We ablate the generator on the full internal-coordinate glycerol target along three independent axes: the loss (forward KL versus the X-regularized $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$); the regularization schedule (*sharpen*: the soft-core cap e_{cap} is geometrically annealed from 100 to 200 kJ/mol along the ladder with a Monte Carlo sharpening step per stage; *raw*: fixed $e_{\text{cap}} = 200$ kJ/mol, no anneal or sharpening); and the wide-coverage

sampler (plain quench and temper versus its δ -reweighted variant, $\delta = 0.1$, defined only for the X loss). The headline is the error-propagation factor F (S20); the *raw* runs take no sharpening step ($\text{ESS}^{\text{sharp}} \equiv 1$).

loss (raw pushforward)	K	wall (min)	F
annealed SMC	10	2	635.56
forward KL	7	215	40.86
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$	6	162	21.96
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$, δ -rw.	6	158	16.56
loss (sharpening)	K	wall (min)	F
annealed SMC	10	5	463.02
forward KL	4	88	44.00
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$	4	73	22.51
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$, δ -rw.	4	73	14.16

Table S9: Glycerol ($d = 36$) ablation, raw pushforward (top) and sharpening (bottom): ladder length K , training wall-clock, and the error-propagation factor F (S20) (smaller better; the sharpening rows include the per-stage Monte Carlo sharpening reweight). δ -rw. = δ -reweighted quench and temper.

The X regularizer is the dominant factor (Table S9), roughly halving F at either schedule. The δ -reweighted quench and temper lowers F further in both schedules ($22.5 \rightarrow 14.2$ under sharpen, $22.0 \rightarrow 16.6$ under raw). Once the sharpening reweight is charged to F , the *sharpen* schedule no longer beats the fixed-cap *raw* run on F alone ($22.5 \approx 22.0$); its advantage is feasibility and speed, reaching $t = 1$ in 4 stages / 73 min versus 6–7 stages / 158–215 min. The lowest- F and fastest run is KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$, δ -reweighted, under the sharpen schedule. Turning the trained flow off entirely (the identity map at every stage, i.e. plain adaptive annealed SMC over the same ladder and acceptance standard) gives $F = 463$ (sharpen) and $F = 636$ (raw), each in $K = 10$ stages. The trained flow thus cuts the propagation factor by ~ 30 – $40\times$ while shortening the ladder from $K = 10$ to $K = 4$ – 6 (Figure S13): this is the quantitative value the transport map adds over annealing alone.

Glycerol is achiral (each torsion marginal is even, $p(\phi) = p(-\phi)$; Figure S15), and every rotatable torsion is three-fold: gauche $^\pm$ near $\pm 60^\circ$ and trans near 180° (Figure S14). The $\pm\phi$ symmetry pairs gauche $^- \leftrightarrow$ gauche $^+$ and leaves trans self-symmetric, so the joint target over the five rotatable torsions is combinatorially multimodal (dozens of conformers), which is what the wide-coverage pool must cover.

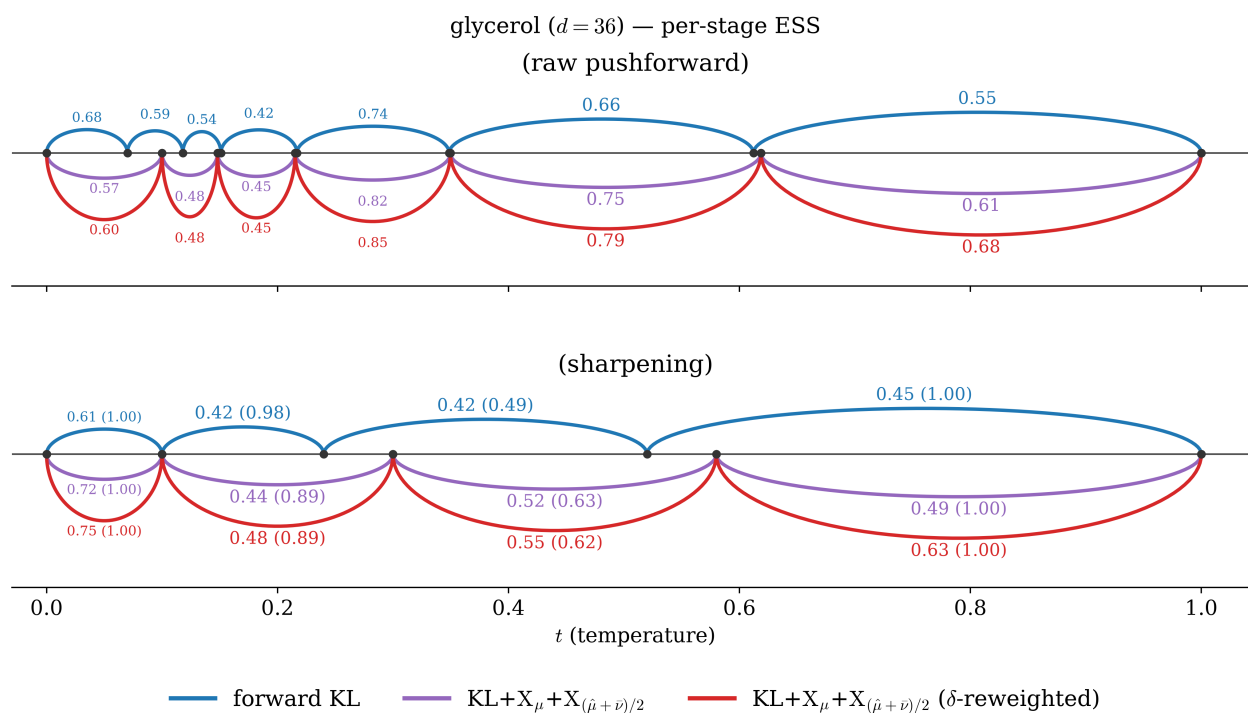


Figure S13: Glycerol per-stage validation ESS as annealing-ladder arcs, raw (top) and sharpen (bottom). In each row forward KL is drawn above the axis (blue), $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ below (purple), and its δ -reweighted variant further below (red). Each arc is an accepted stage; the apex label is that stage's validation ESS; a cross marks a bridge the ladder failed to reach.

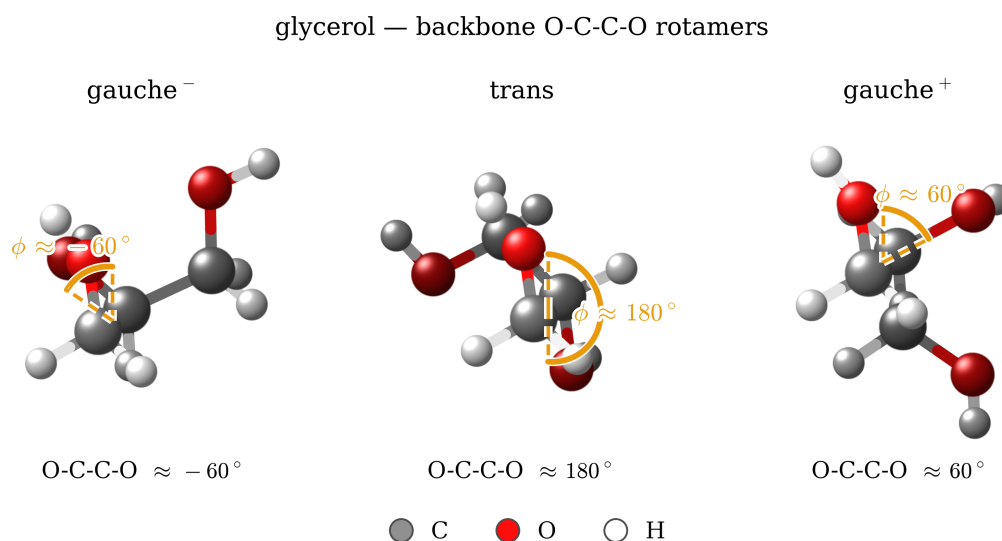


Figure S14: Typical glycerol configurations at the three states of the backbone O-C-C-O torsion: gauche[−] (-60°), trans (180°), gauche⁺ ($+60^\circ$). The gold arc marks the dihedral ϕ ; ball-and-stick (C grey, O red, H white).

glycerol ($d = 36$) — torsion marginals: BG vs annealed SMC reference

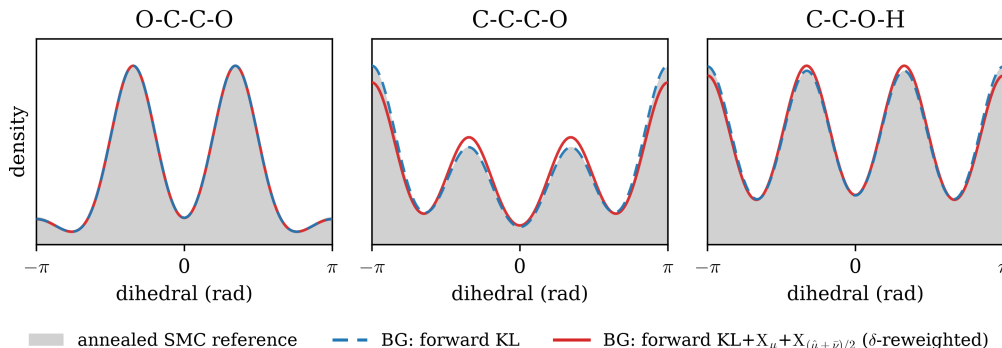


Figure S15: Glycerol torsion marginals of both generators against a converged annealed SMC reference (grey): forward KL (blue) and $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted (red), both sharpened with the cap annealed to $e_{\text{cap}} = 200$ kJ/mol. Both recover the multimodal gauche $^\pm$ /trans structure of the characteristic glycerol torsions.

S3.4.2 Diethanolamine ($d = 48$)

We test the generator on the full internal-coordinate diethanolamine target ($\text{HN}(\text{CH}_2\text{CH}_2\text{OH})_2$, $d = 48$), comparing the two best losses under the same sharpening schedule (the soft cap geometrically annealed $e_{\text{cap}} : 100 \rightarrow 200$ kJ/mol, a Monte Carlo sharpening step per stage): forward KL, and the X-regularized $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ with the δ -reweighted quench and temper ($\delta = 0.1$). Forward KL reaches $t = 1$ in $K = 6$ stages, the X-regularized loss in $K = 5$, a single direct $0.65 \rightarrow 1$ step.

loss	K	wall (min)	F
annealed SMC	10	7	1455.44
forward KL	6	896	92.60
$\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -rw.	5	782	35.90

Table S10: Diethanolamine ($d = 48$): two trained-flow losses and an annealed SMC reference, all with a per-stage Monte Carlo sharpening reweight (cap e_{cap} geometrically annealed $100 \rightarrow 200$ kJ/mol): ladder length K , training wall-clock, and the error-propagation factor F (S20) (smaller better; F includes the sharpening reweight). δ -rw. = δ -reweighted quench and temper.

The X-regularized loss with δ -reweighting reaches $F = 35.9$ versus forward KL’s $F = 92.6$ (Table S10), a $\sim 2.6\times$ reduction in importance-weight waste at identical settings (glycerol showed a comparable $\sim 3.1\times$ gap, 14.2 vs 44.0). The advantage shows in the late, sharp stages and in the ladder length: forward KL’s per-stage validation ESS stalls at 0.46–0.53 and it needs 6 rungs to reach $t = 1$, whereas the $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted run holds a higher ESS (0.74 on the final, direct $0.65 \rightarrow 1$ step) and reaches $t = 1$ in 5 (Figure S16). This is a singular, high-dimensional, combinatorially multimodal target, so each run is a multi-hour computation; closing it efficiently is the practical contribution of the X-regularized loss.

Diethanolamine has a plane of symmetry (each torsion marginal is even, $p(\phi) = p(-\phi)$); its rotatable backbone torsions, O-C-C-N and C-C-N-C along the ethanolamine arms plus the hydroxyl rotations, are each multi-well (gauche $^\pm$ /trans, Figure S17), so the joint target is combinatorially multimodal, which the wide-coverage pool must cover.

Against the annealed SMC reference (Figure S18) the two losses are visually indistinguishable: both recover the multimodal backbone torsions and neither is discernibly closer to the reference. The separation is in efficiency, not in the sampled marginals: $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ reaches $t = 1$ in fewer stages (5 vs 6) with a higher final-step ESS (0.74 vs 0.53), a $\sim 2.6\times$ smaller error-propagation factor ($F = 35.9$ vs 92.6), and fewer rejected stage attempts in the early ladder (stages 2–4: 1, 1, 0 rejections versus forward KL’s 2, 2, 1). Switching the trained flow off entirely (the identity map at every stage, i.e. plain adaptive annealed SMC

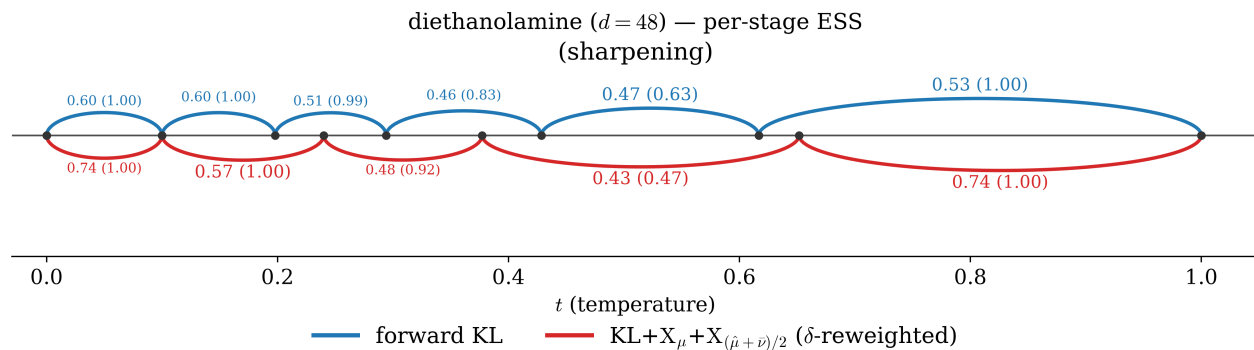


Figure S16: Diethanolamine per-stage ESS as annealing-ladder arcs (sharpening schedule): forward KL above the axis (blue), $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted, below (red). Each arc is an accepted stage; the apex label is that stage's validation ESS, with the sharpening ESS in parentheses.

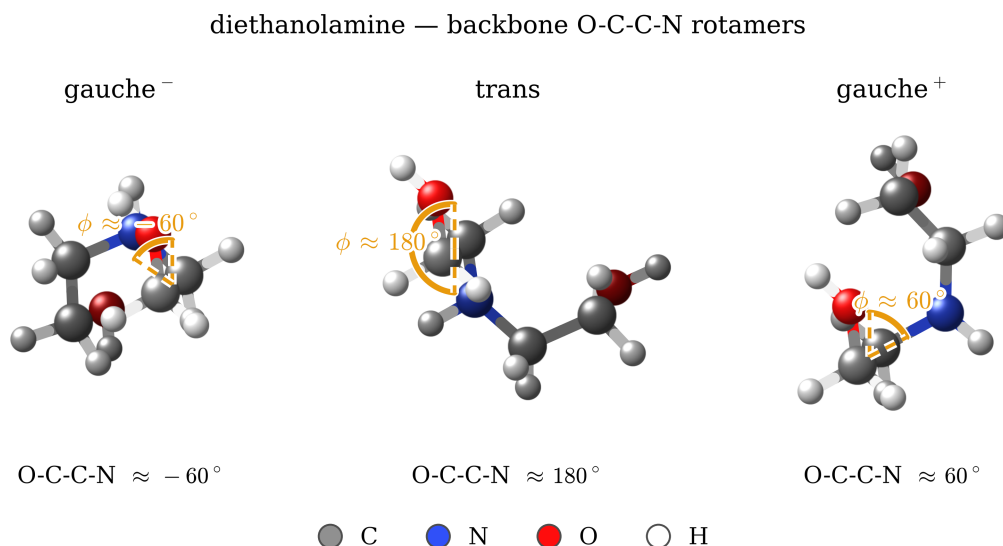


Figure S17: Typical diethanolamine configurations at the three states of a backbone O-C-C-N torsion: gauche⁻ (-60°), trans (180°), gauche⁺ ($+60^\circ$). The gold arc marks the dihedral ϕ ; ball-and-stick (C grey, N blue, O red, H white).

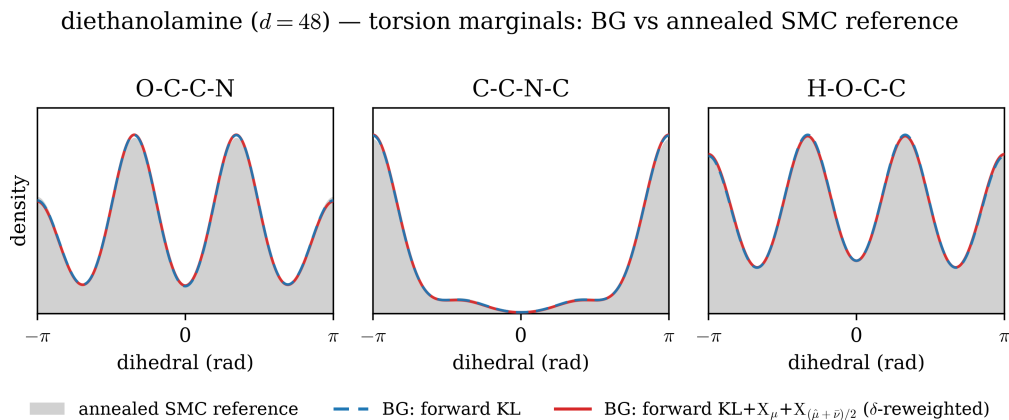


Figure S18: Diethanolamine torsion marginals of both generators against a converged annealed SMC reference (grey): forward KL (blue) and $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted (red), both sharpened with the cap annealed to $e_{\text{cap}} = 200$ kJ/mol. Both recover the multimodal backbone torsions and track the annealed SMC reference.

over the same ladder) gives $F = 1455$ in $K = 10$ stages. The trained X-regularized flow therefore cuts the propagation factor by $\sim 40\times$ and halves the ladder length (5 vs 10): on this singular target the transport map adds even more over annealing alone than on glycerol, a justification for training a flow.

S3.4.3 Alanine dipeptide ($d = 60$)

We push the generator to alanine dipeptide (Ac-Ala-NHMe), the canonical Boltzmann generator benchmark, in full internal coordinates ($d = 60$). The qualitative step up from glycerol and diethanolamine is the target’s character: those are dominated by smooth rotatable-bond multimodality, whereas alanine dipeptide folds the backbone tightly enough that the Lennard-Jones clash core, not the torsion profile, sets the hardest part of the anneal. We report the X-regularized loss only (KL+ $X_\mu + X_{(\hat{\mu}+\bar{\nu})/2}$, δ -reweighted, $\delta = 0.1$), with a per-stage Monte Carlo sharpening in which *both* the cap $e_{\text{cap}}: 100 \rightarrow 200$ kJ/mol and the clash floor $r_{\text{floor}}: 0.2 \rightarrow 0.1$ nm are annealed arithmetically, unlike the geometric e_{cap} schedule of glycerol and diethanolamine; alanine dipeptide is also the only target that anneals r_{floor} , which glycerol and diethanolamine hold fixed at 0.1 nm. The run reaches $t = 1$ in $K = 11$ stages at $F^* = 67.95$, with every stage holding validation ESS ≥ 0.51 and sharpening ESS ≥ 0.95 (Table S11, Figure S19). The bare composed flow’s end-to-end ESS against the sharp target, from a single importance reweight of source samples pushed through the fully composed K -stage map and scored against $U_{1,1}$, is only $\sim 5 \times 10^{-6}$, so the per-stage sharpening, not a raw pushforward, is the deliverable.

loss	K	wall (min)	F^*
annealed SMC	12	23	129.70*
KL+ $X_\mu + X_{(\hat{\mu}+\bar{\nu})/2}$, δ -rw.	11	~ 1000	67.95*

Table S11: Alanine dipeptide ($d = 60$), the X-regularized loss with a per-stage Monte Carlo sharpening reweight (the cap e_{cap} and floor r_{floor} annealed arithmetically): ladder length K , wall-clock, and the error-propagation factor F^* (S22), starred throughout because every entry uses the smooth validation ESS (smaller better; F^* includes the sharpening reweight). The first row is an annealed SMC reference, the identity map over the same ladder. δ -rw. = δ -reweighted quench and temper.

For comparison, the lighter targets close at $F = 14.2$ (glycerol) and $F = 35.9$ (diethanolamine) with the same loss; alanine dipeptide’s larger F^* is the price of its singular nonbonded core, not of its multimodality. Three alanine-dipeptide-only factors nonetheless make this F^* read smaller than a strict like-for-like extension of that trend: the per-stage ESS is the smooth estimator $\text{ESS}_k^{\text{val}*}$ (S21) rather than the raw weighted ESS; r_{floor} is annealed alongside the cap; and the ladder is much longer ($K = 11$ versus 4 and 5), its smaller per-step jumps keeping each per-stage ESS high. All three raise the per-stage ESS entering F^* , and none applied to the lighter targets.

The instability is sharply localized to a clash bottleneck near $t \approx 0.3$ –0.43. The stages spanning $t \approx 0.29$ –0.43 (soft cap $e_{\text{cap}} \approx 129$ –143 kJ/mol) are where the nonbonded core first bites: the run minimum validation ESS (0.51 at $t = 0.39$) sits here, and these are the *only* stages that shrink the step (shrunk once at stages 3, 4, 5) or retry heavily (3 attempts at stages 5, 6). Below $t \approx 0.3$ the torsions still dominate and the flow trains in a single attempt; above $t \approx 0.5$ the cap is sharp enough that the basins separate cleanly and ESS recovers to ≈ 0.8 . The binding constraint for alanine dipeptide is the clash singularity, not the multimodality: the qualitative difference from the $d \leq 48$ molecules, and the reason the three devices of Section S3.3 (particle drop, best ESS gate snapshots, r -floor anneal), inert on glycerol and diethanolamine, were necessary here.

The per-stage ESS history (Figure S19) exposes two coupled instabilities that the best ESS gate works around but does not cure. First, the within-stage ESS is *non-monotone and unstable*: it can rise, plateau, then drop abruptly (the sharp collapse late in the $t = 0.65$ stage and the volatility at $t = 1$ are the clearest cases), so the last training step is frequently far below the best (stage 9’s training ESS decays to 0.03 by step 500 while the best ESS gate keeps a snapshot at validation ESS = 0.80). Second, near the clash bottleneck ($t \approx 0.3$ –0.5) several flows *degenerate toward the identity map*: the best achievable ESS barely exceeds the $t = 0$ identity baseline ($\Delta = \text{best} - \text{identity} \approx 0$ at stages 3, 4, 7), i.e. training fails to improve on doing nothing. Both are understood consequences of the *singularity of the potential*: the Lennard-Jones $r \rightarrow 0$ divergence leaves the soft-capped target ill-conditioned exactly where the nonbonded core first bites, so the

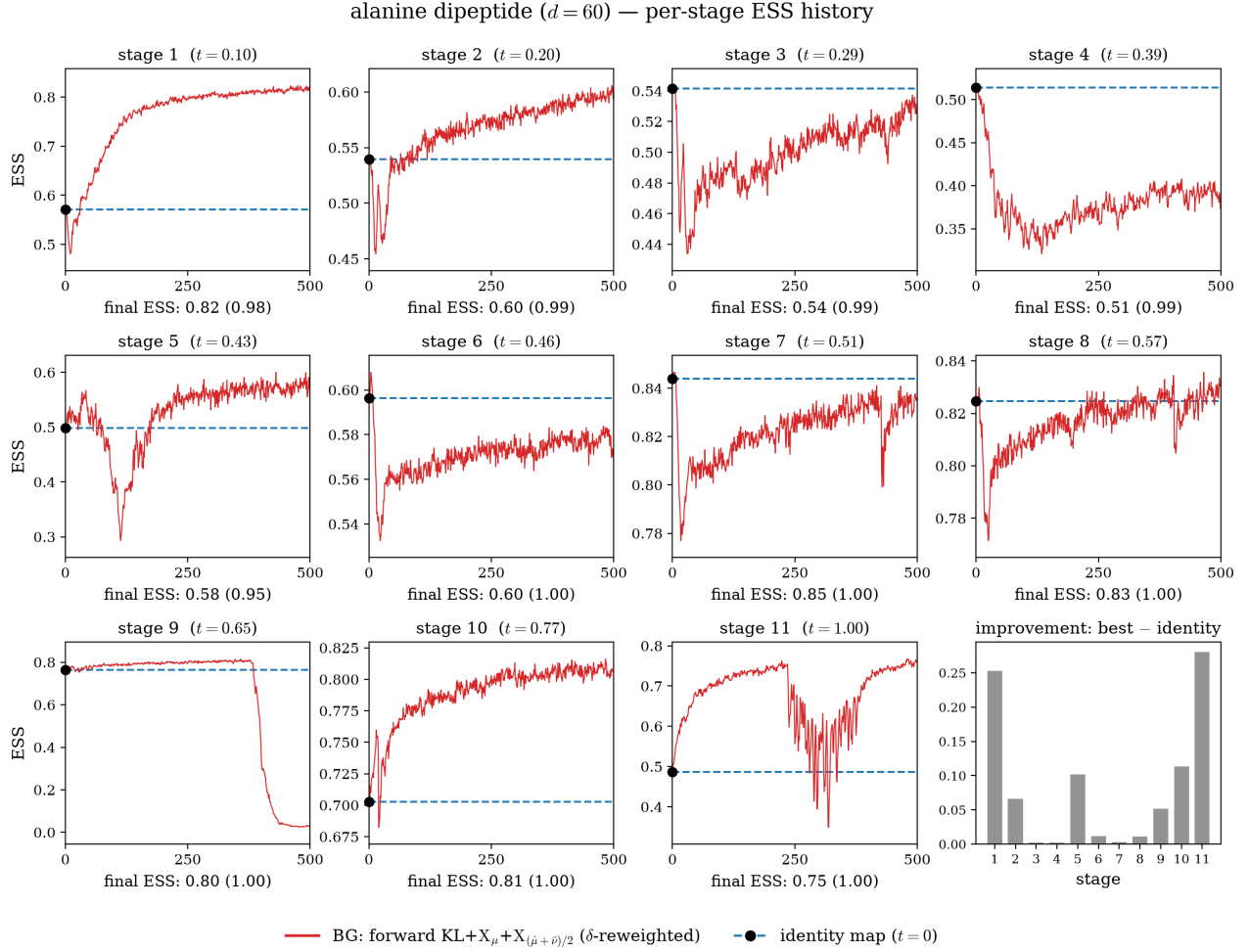


Figure S19: Alanine dipeptide per-stage ESS history: each panel is the per-step effective sample size during that stage’s X-regularized δ -reweighted flow training (red), against the identity-map baseline at $t = 0$ (blue dashed). Below each panel is that stage’s accepted validation ESS with the sharpening ESS in parentheses. The dip across $t \approx 0.29$ – 0.43 is the clash bottleneck, with ESS recovering to ≈ 0.8 once the cap is sharp enough to separate the basins.

gradient signal that would pull the flow off identity is weakest there. We report this as a known limitation; robust remedies (stronger or adaptive regularization annealing, parameterizations tailored to the repulsive wall) are left to future work.

alanine dipeptide ($d = 60$) — torsion marginals: BG vs annealed SMC reference

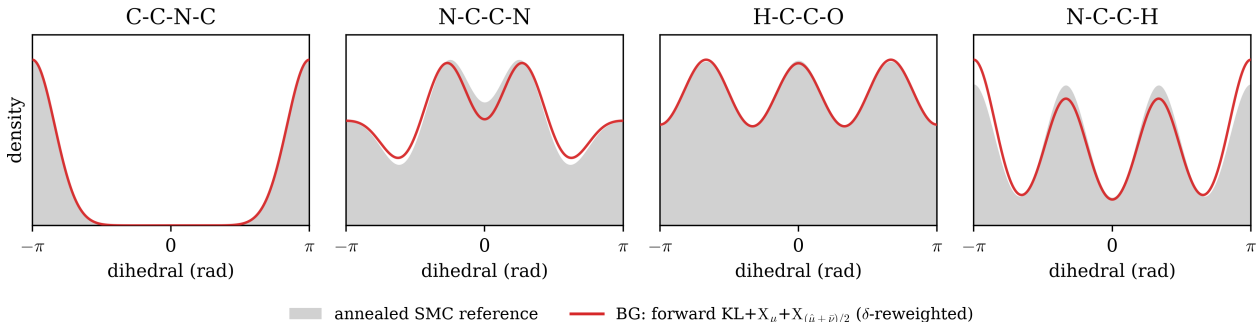


Figure S20: Alanine dipeptide torsion marginals of the BG deliverable (red) against a converged annealed SMC reference (grey). The four most multimodal proper torsions are shown; the densities are $\pm\phi$ -symmetrised, the achiral target having $p(\phi) = p(-\phi)$. The BG recovers the multimodal backbone torsions and tracks the reference; the peptide-bond torsion ω (C-C-N-C) is trans-dominant in the generator, the one torsion whose high rotation barrier also makes the uniform-init annealed SMC reference slow to converge.

alanine dipeptide’s backbone ϕ, ψ are multi-basin (Figure S20), and a second, alanine-dipeptide-specific feature is chirality: the classical force field is mirror-symmetric (every term in (S17) is invariant under reflection), so the L- and D-forms are isoenergetic. A physical MD trajectory stays in one form, as inverting the C_α needs bond breaking, but the internal-coordinate flow carries no chirality constraint and emits *both* (Figure S21). In the Ramachandran (Figure S22) this appears as the $(\phi, \psi) \rightarrow (-\phi, -\psi)$ point symmetry: the D-basins are the exact mirror of the L-basins, each populated, where a single kinetically trapped physical trajectory would show only one.

alanine dipeptide — L and D enantiomers

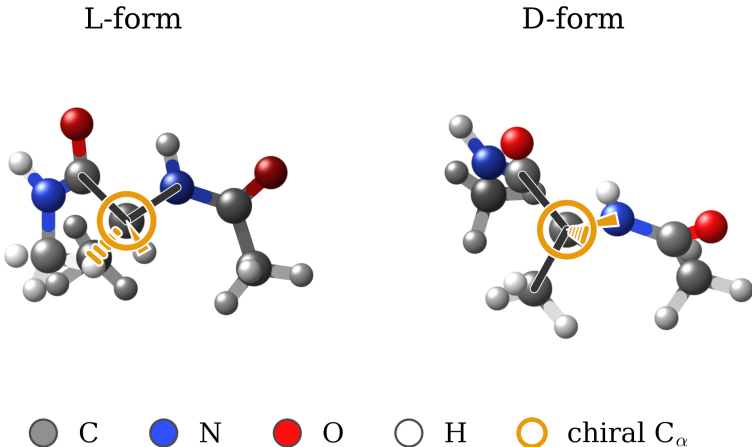


Figure S21: The two enantiomers of alanine dipeptide: the energy-minimized L-form and the D-form, a 3D reflection of it, shown from different viewpoints (ball-and-stick; C grey, N blue, O red, H white). The chiral C_α is ringed in gold with stereo bonds, a solid wedge toward the viewer and hashed bonds behind. The achiral Amber force field makes the two isoenergetic, so the generator emits both, the structural companion to the $(\phi, \psi) \rightarrow (-\phi, -\psi)$ Ramachandran symmetry of Figure S22.

The X-regularized BG closes the full internal-coordinate alanine dipeptide target to $t = 1$ in 11 stages

alanine dipeptide ($d = 60$) — Ramachandran: BG vs annealed SMC reference

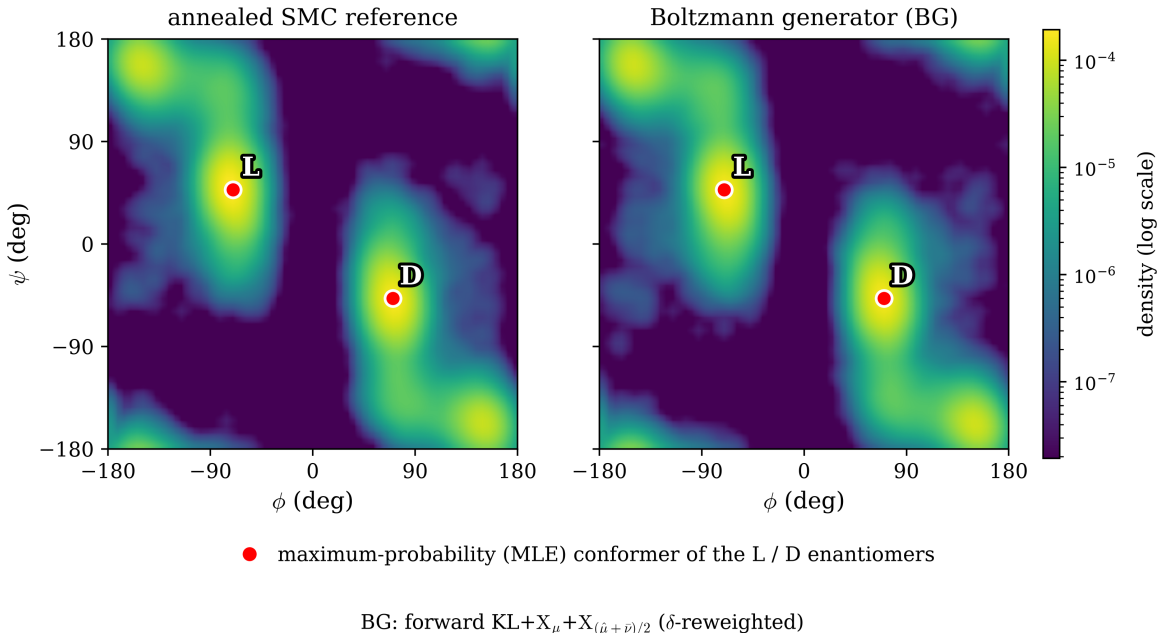


Figure S22: Alanine dipeptide Ramachandran (ϕ, ψ), shared log-density scale: the BG deliverable (right) against the annealed SMC reference (left). Red dots mark the maximum-probability conformer of each enantiomer (L at $\phi < 0, \psi > 0$; D at $\phi > 0, \psi < 0$). Both panels populate the same backbone basins with the $(\phi, \psi) \rightarrow (-\phi, -\psi)$ point symmetry: the achiral vacuum force field makes the L- and D-enantiomers isoenergetic, and the generator, carrying no chirality constraint, samples both (Section S3.4.3).

at $F^* = 67.95$, recovering the multimodal backbone torsions against a converged annealed SMC reference and emitting both chiral forms of the mirror-symmetric target. The lesson carried over from the lighter $d \leq 48$ molecules is inverted: alanine dipeptide’s difficulty is not its combinatorial torsion multimodality but its nonbonded clash core, concentrated in the $t \approx 0.3$ – 0.43 stages, which the particle drop, best ESS gate snapshots, and r -floor anneal together make tractable. Across all three molecules the pattern is consistent: the sharpening split makes the singular target trainable at all, and the X-regularized flow then earns a ~ 30 – $40\times$ reduction in importance-weight waste where the landscape is multimodal but clash-mild, narrowing to $\sim 2\times$ where the singularity itself dominates.

S3.5 Limitations

Four deliberate simplifications bound the molecular experiments, each a direction for future work. First, the targets are evaluated in vacuum: U_{FF} is the isolated-molecule Amber potential (S17) with no solvent and no nonbonded cutoff. Solvation only reshapes U_{FF} and not the sampler, and the vacuum targets already display the well-separated potential wells the generator must discover and resolve, but implicit or explicit solvent and larger peptides are left untreated. Second, the internal-coordinate generator carries no chirality constraint, so for the achiral Amber force field it samples both the L- and D-enantiomers (Section S3.4.3), whereas a physical trajectory stays in one; isolating a single enantiomer by a filter or restraint is left to future work. Third, we do not benchmark against FAB (Midgley et al.): its large per-step batches and replay buffer make a like-for-like comparison under the same compute budget as these forward KL-type losses prohibitive, so the natural same-cost baselines here are the bare forward KL and the identity-flow annealed SMC. Fourth, on the alanine dipeptide clash core the per-stage flow training is unstable and can degenerate toward the identity map, as documented in Section S3.4.3; the best ESS gate works around this but does not cure it, and a regularization or parameterization that conditions the repulsive wall remains open.

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